CHAPTER 10

# Nitrogen- and Phosphorus-Containing Compounds

## Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: 24794835489d402ee6a6d13247d03839e0514f1e95103787f9b8171508a3078a

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

### Coverage checklist

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ch10:molecule:alanine | alanine

ch10:molecule:glycine | glycine

ch10:molecule:glycylglycine | glycylglycine

ch10:molecule:aminonitrile | aminonitrile

ch10:molecule:phthalimide-anion | phthalimide-anion

ch10:molecule:dimethylphosphate | dimethylphosphate

ch10:molecule:phosphoricacid | phosphoricacid

ch10:curriculum | scope-1, scope-2, scope-3

# Curriculum scope

scope-1: 10.1 Amino Acids, Peptides, and Proteins

scope-2: 10.2 Synthesis of α-Amino Acids

scope-3: 10.3 Phosphorus-Containing Compounds

Objective: Analyze amino-acid charge and stereochemistry

Objective: Identify peptide bonds and synthesis carbon sources

Objective: Distinguish phosphate linkages and reaction energetics

Supplied Kaplan curriculum: Chapter 10; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter10b.xhtml

Buffer action and capacity: OpenStax Chemistry 2e 14.6, inspected | https://openstax.org/books/chemistry-2e/pages/14-6-buffers

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

# amino-acids | Amino acids, charge, and stereochemistry

Ionization depends on pH

## amino-acids:rule | Core Rules

An alpha amino acid has amino and carboxyl groups attached to the same alpha carbon. At many near-neutral pH conditions, simple amino acids predominantly form zwitterions. Side chains add nonpolar, aromatic, polar, acidic, or basic behavior. Glycine has no stereocenter; most proteinogenic L amino acids are S, with cysteine the familiar R exception. Amino acids are amphoteric: an acidic carboxyl group can donate H+ and a basic amino group can accept H+; side chains may add further ionizable sites.

Visual card: Amino Acid Zwitterions, Titration Curves & Isoelectric Point (pI) Formulas

Chemistry diagram: Illustrative amino-acid zwitterions. Formal charges can sum to net zero.

Neutral R: pI = ½(pKa_COOH + pKa_NH3+) | Acidic (Asp/Glu): pI = ½(pKa_COOH + pKa_R) | Basic (Lys/Arg/His): pI = ½(pKa_R + pKa_NH3+)

pKa(α-COOH) ~2.0 | pKa(α-$NH_3^+$) ~9.5

Zwitterion at pH 7.4 (+1/−1 = 0 Net)

pI = Average of 2 Flanking pKas!

pH < 2 (Strong Acid) | Fully protonated (+1 or +2 if basic R) | $+H_3N–CH(R)–COOH$ (moves to cathode −)

pH = pI (Isoelectric Point) | Net charge = 0.0 (Zwitterion) | $+H_3N–CH(R)–COO^-$ (minimum solubility; zero migration)

pH > 10 (Strong Base) | Fully deprotonated (−1 or −2 if acidic R) | $H_2N–CH(R)–COO^-$ (moves to anode +)

Why Never Average All 3 pKa Values! — — Always find the neutral Zwitterion species (net charge = 0) first, then average ONLY the two pKa steps that immediately create and destroy that 0-charge form!

Key Side-Chain pKa Values to Memorize — — Asp (~3.7), Glu (~4.3), His (~6.0, near physiological pH!), Cys (~8.3), Tyr (~10.1), Lys (~10.5), Arg (~12.5).

## amino-acids:cue | Memory Cues

Low pH adds protons; high pH removes them. Net charge is the sum of all ionized groups.

Visual card: Instant Amino Acid Charge & Isoelectric Point (pI) Calculator

Which amino acid class are you calculating pI or charge for at pH 7.4?

Nonpolar / Uncharged Polar (Gly, Ala, Val, Leu, Ile, Ser, Thr, Asn, Gln) — pI ≈ ½(2.2 + 9.5) ≈ 5.8–6.0 | Net Charge at pH 7.4 = 0 — Only two ionizable groups exist: α-$COO^-$ (−1) and α-$NH_3^+$ (+1) cancel to zero net charge at pH 7.4.

Acidic Side Chains: Aspartate (D) & Glutamate (E) — pI = ½(pKa1_αCOOH + pKa2_R) ≈ 3.0 | Net Charge at pH 7.4 = −1 — Neutral zwitterion sits between the two acidic deprotonations (around pH 3); at pH 7.4 both carboxylates are −1 plus one +1 α-$NH_3^+$ = −1 net.

Basic Side Chains: Lysine (K), Arginine (R) & Histidine (H) — pI = ½(pKa2_R + pKa3_αNH3+) ≈ 9.7–10.8 (His ≈ 7.6) | Charge at pH 7.4: Lys/Arg = +1, His ≈ 0 — Neutral zwitterion requires deprotonating α-COOH (−1) AND one of the two ammonium groups (+1); average the two higher pKa values!

## amino-acids:trap | Common Mistakes / MCAT Traps

A zwitterion has formal charges despite net zero charge. The pI is the pH of average net zero charge; for ionizable side chains, average the two pKa values bracketing the neutral species, not automatically the terminal pKa pair.

Visual card: Drawing Uncharged $H_2N–CH(R)–COOH$ in Water or Averaging All 3 pKa Values for pI

Common Trap — — Averaging (2.0 + 3.9 + 9.8)/3 for Aspartate's pI, or drawing an amino acid with neutral $–NH_2$ and neutral –COOH in the same molecule.

Correct Reasoning — — Because –COOH (pKa ~2) is a billion times more acidic than $–NH_3^+$ (pKa ~9.5), –COOH ALWAYS loses its proton long before $–NH_3^+$ does (forming the dipolar zwitterion)! Average only the 2 pKas flanking the 0-charge species.

## amino-acids:app | MCAT Applications

Aliphatic side chains such as Ala, Val, Leu, and Ile are nonpolar; Ser and Thr have polar OH groups; Asn and Gln have polar amides. Asp/Glu are usually negatively charged near neutral pH; Lys/Arg usually positive. Histidine protonation is strongly pH- and environment-dependent. Aromatic does not always mean purely nonpolar: Tyr has a phenolic OH. At pH below pI a protein tends positive; above pI it tends negative. Methionine has a largely nonpolar thioether side chain; cysteine has a polar thiol and can form disulfides on oxidation. Proline has a cyclic side chain bonded to its nitrogen (a secondary amine in the free amino acid); phenylalanine and tryptophan are commonly grouped as nonpolar aromatic amino acids, although tryptophan has an indole N–H donor.

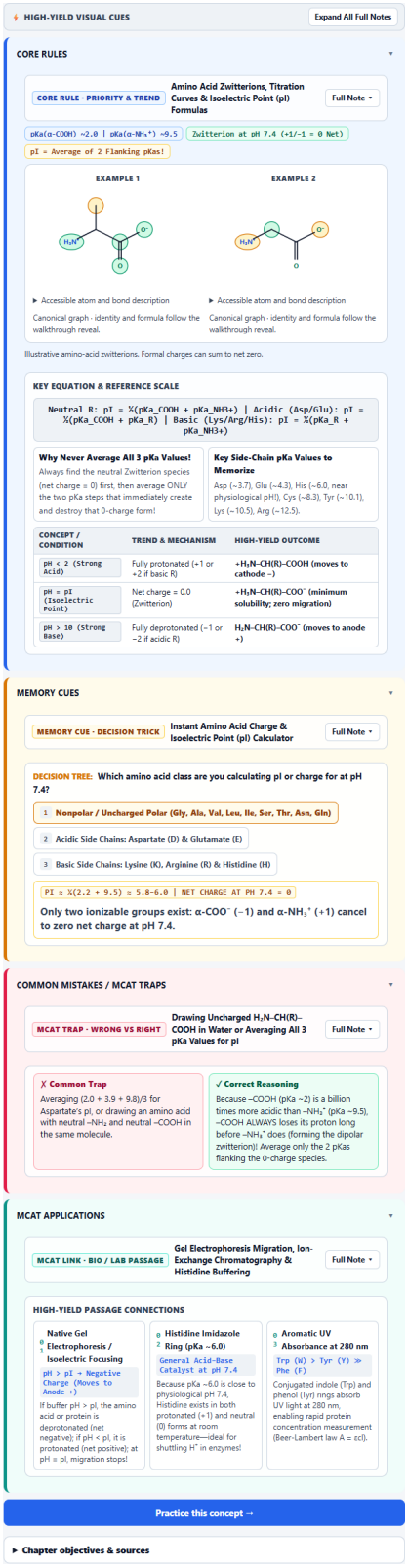
Visual card: Gel Electrophoresis Migration, Ion-Exchange Chromatography & Histidine Buffering

Native Gel Electrophoresis / Isoelectric Focusing — pH > pI → Negative Charge (Moves to Anode +) — If buffer pH > pI, the amino acid or protein is deprotonated (net negative); if pH < pI, it is protonated (net positive); at pH = pI, migration stops!

Histidine Imidazole Ring (pKa ~6.0) — General Acid–Base Catalyst at pH 7.4 — Because pKa ~6.0 is close to physiological pH 7.4, Histidine exists in both protonated (+1) and neutral (0) forms at room temperature—ideal for shuttling $H^+$ in enzymes!

Aromatic UV Absorbance at 280 nm — Trp (W) > Tyr (Y) ≫ Phe (F) — Conjugated indole (Trp) and phenol (Tyr) rings absorb UV light at 280 nm, enabling rapid protein concentration measurement (Beer-Lambert law $A = \varepsilon cl$).

Screenshot: amino-acids-visuals



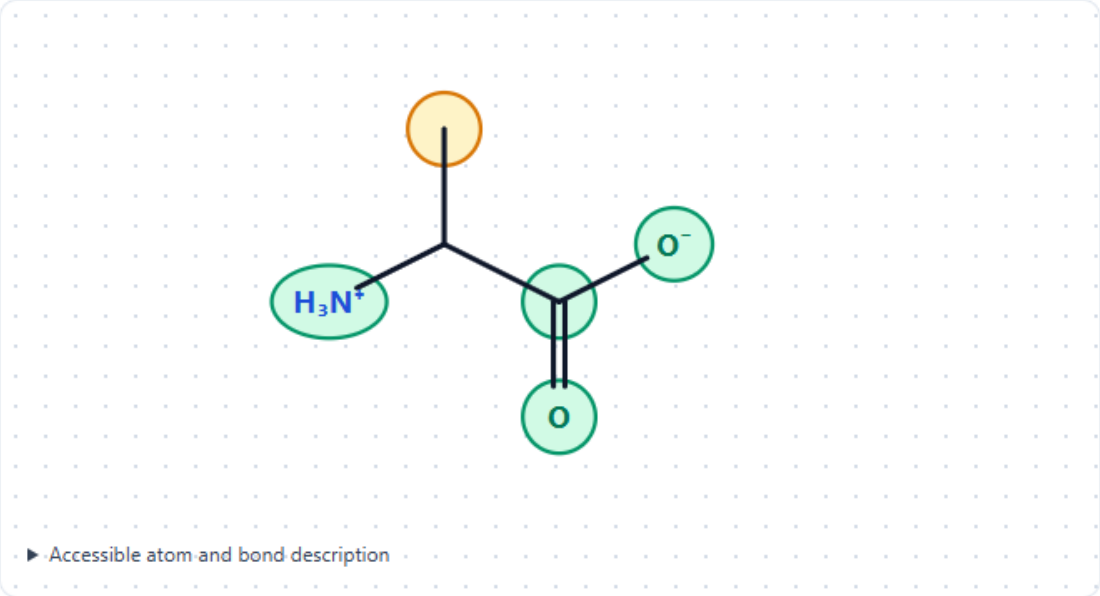

## amino-acids:step-1 | 1. Zwitterionic alanine

The nitrogen bears positive charge and the carboxylate bears negative charge. Its alpha carbon also has methyl and H. This connectivity drawing leaves stereochemistry unspecified.

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Screenshot: amino-acids-step-1



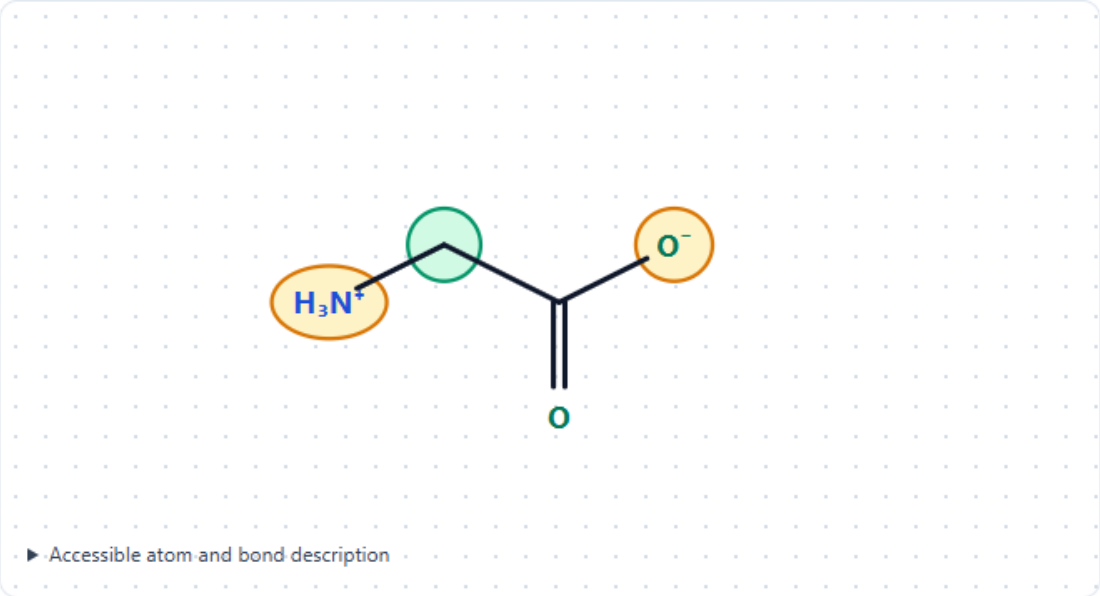

## amino-acids:step-2 | 2. Glycine exception

Glycine has two hydrogens at its alpha carbon and is achiral. Its neutral net charge near suitable pH does not mean it lacks charged sites.

Displayed molecule ID: glycine

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Screenshot: amino-acids-step-2




## amino-acids:step-3 | 3. Relative configuration

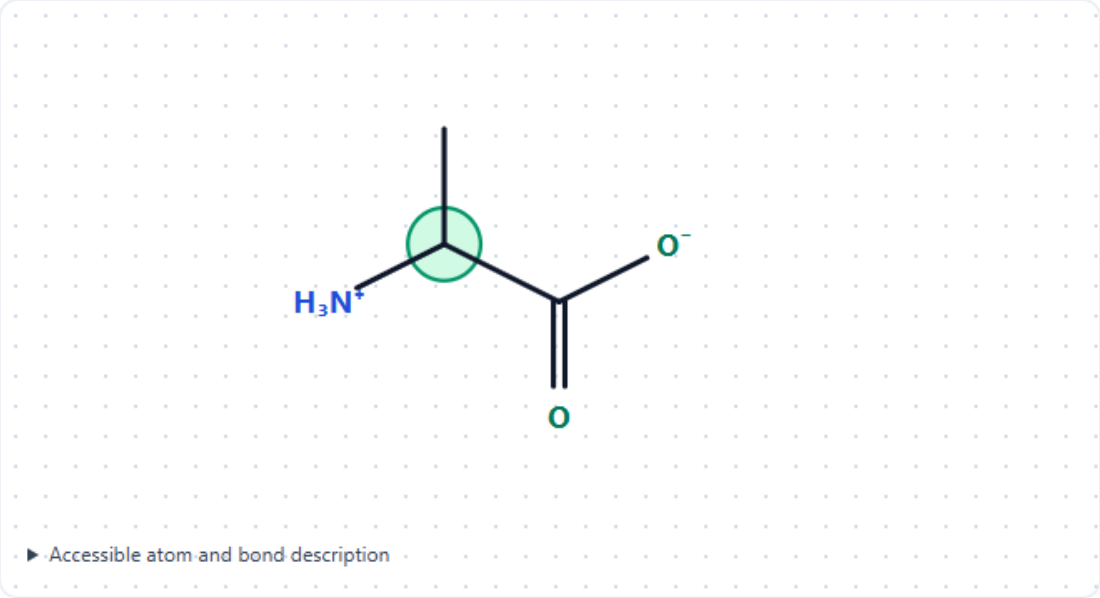
The separate Fischer projection below shows L-alanine: carboxyl above, methyl below, amino left. The skeletal canvas above specifies connectivity without configuration. In the Fischer projection, horizontal bonds point toward you. L does not specify

the optical-rotation sign.

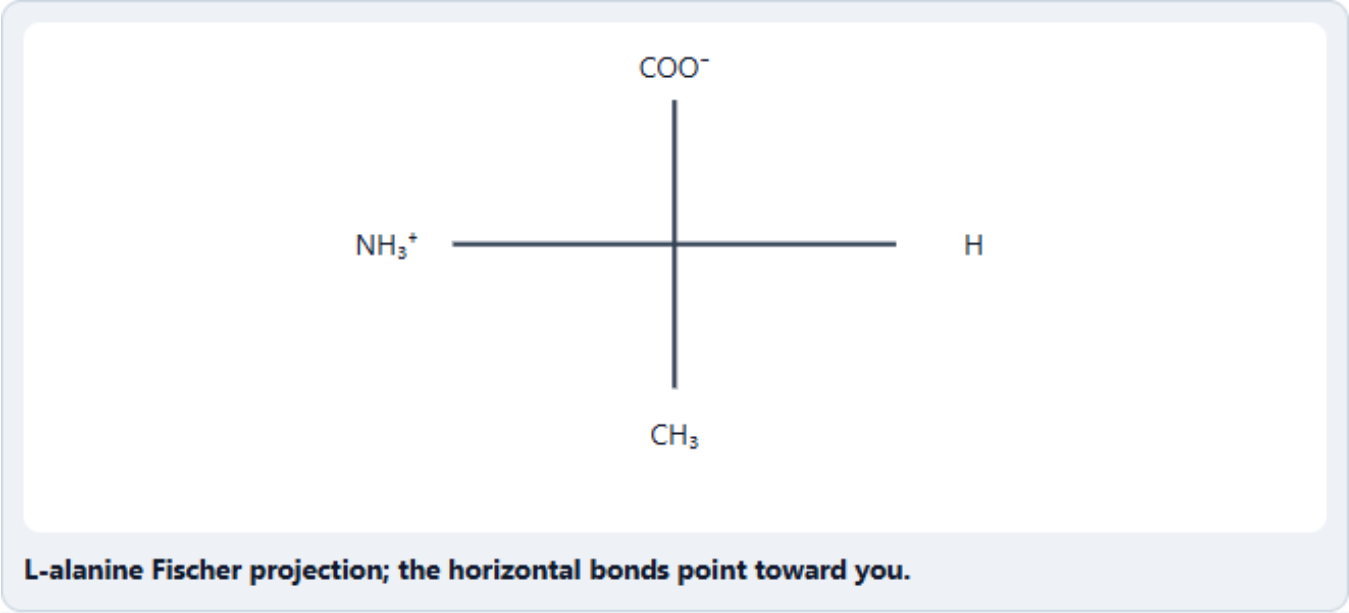
Displayed molecule ID: alanine

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Screenshot: amino-acids-step-3




Screenshot: amino-acids-step-3-diagram




{"diagramSpecification": {"type": "fischer", "top": "COO⁻", "bottom": "CH₃", "left": "NH₃⁺", "right": "H", "caption": "L-alanine Fischer projection; the horizontal bonds point toward you."}}

# peptides | Peptide bonds and protein structure

An amide links amino-acid residues

## peptides:rule | Core Rules

A peptide bond joins one residue's carbonyl carbon to the next residue's nitrogen. Resonance makes the peptide group approximately planar with restricted C–N rotation. Sequence is conventionally read N-terminus → C-terminus. Primary sequence supports secondary structures, tertiary folding, and sometimes quaternary assembly. Net peptide formation is a condensation/dehydration, eliminating water from amino and carboxyl groups in formal bookkeeping. Activation enables synthesis; hydrolysis consumes water to cleave the amide, with enzymes or suitable strong acid/base and often heat. Basic cleavage is base-promoted and gives carboxylate, not necessarily a neutral acid before workup.

Visual card: Peptide Bond Resonance Planarity, N→C Convention & Protein Structure Levels (1°–4°)

Chemistry diagram: Locate the peptide C–N linkage and the termini. The drawing is not a measured three-dimensional geometry.

Write N-Terminus ($+H_3N$) → C-Terminus ($COO^-$)

Peptide Bond = $sp^2$ Planar (Trans)

1°=Sequence, 2°=α/β Backbone H-Bonds, 3°=Side Chains, 4°=Subunits

1° Structure (Primary) | Linear amino acid sequence (N → C) | Covalent peptide (amide) bonds; cleaved by 6M HCl/proteases

2° Structure (Secondary) | α-Helix (i to i+4) & β-Sheet | Backbone C=O···H–N H-bonds ONLY (Proline = helix breaker!)

3° / 4° Structure | 3D fold (3°) & Multi-subunit complex (4°) | Hydrophobic core, salt bridges, H-bonds, Cys–S–S–Cys disulfides

Directional N-Terminus → C-Terminus Convention — — Amide resonance gives partial C–N double-bond character and a nearly planar peptide group. Trans is usually favored; cis populations depend on the residue and environment.

Partial Double-Bond Character of the Peptide Bond — — Resonance (O=C–NH ↔ $^-$O–C=$NH^+$) gives the C–N bond ~40% π character (1.32 Å vs 1.47 Å), locking all 6 peptide-plane atoms in a rigid TRANS planar conformation (except ~6% cis before Proline).

Denaturation vs. Hydrolysis — — Urea, guanidinium HCl, heat, or pH extremes disrupt noncovalent 2°/3°/4° interactions (and β-mercaptoethanol/DTT reduces Cys–S–S–Cys disulfides), leaving the covalent 1° peptide sequence completely intact!

## peptides:cue | Memory Cues

N to C is the reading direction. Peptide C–N is the restricted bond; adjacent backbone single bonds supply much of the flexibility.

Visual card: Protease Cleavage Specificity & Protein Structure Disrupter Guide

Which reagent or enzyme is acting on the protein?

Trypsin vs. Chymotrypsin vs. Cyanogen Bromide (CNBr) — Trypsin = C-side of Lys/Arg; Chymotrypsin = C-side of Phe/Tyr/Trp; CNBr = C-side of Met — Wait: Proline immediately after the target residue blocks trypsin/chymotrypsin cleavage! The final fragment with NO cleavage site at its C-end is the original C-terminus.

Urea / Heat vs. β-Mercaptoethanol (BME) / DTT — Urea Breaks H-Bonds (2°/3°); BME Reduces Disulfide Bridges (Cys–S–S–Cys → 2 Cys–SH) — In SDS-PAGE, non-reducing vs. reducing gels reveal whether subunits are linked by covalent disulfide bridges!

Proline & Glycine in α-Helices — Both Disrupt α-Helices ("Helix Breakers"), Found in β-Turns & Collagen! — Proline is a secondary amine (lacks a backbone N–H H-bond donor and has a rigid ring); Glycine is too conformationally flexible (high ΔS cost to fold).

## peptides:trap | Common Mistakes / MCAT Traps

A peptide bond is not a C–C bond between two alpha carbons. Peptide formation is thermodynamically and kinetically enabled by activation in cells; plain amino acids in water do not rapidly polymerize by themselves.

Visual card: Thinking Denaturation Breaks Peptide Bonds or Saying Secondary Structure Uses Side-Chain H-Bonds

Common Trap — — Saying boiling an egg or adding 8M urea hydrolyzes peptide bonds, or saying α-helices are held together by R-group interactions.

Correct Reasoning — — Secondary structure (α-helix and β-sheet) is stabilized EXCLUSIVELY by BACKBONE amide C=O···H–N hydrogen bonds; side-chain R-group interactions define tertiary (3°) structure!

## peptides:app | MCAT Applications

Hydrogen bonding organizes alpha helices and beta sheets; hydrophobic effects and side-chain interactions help stabilize folded proteins.

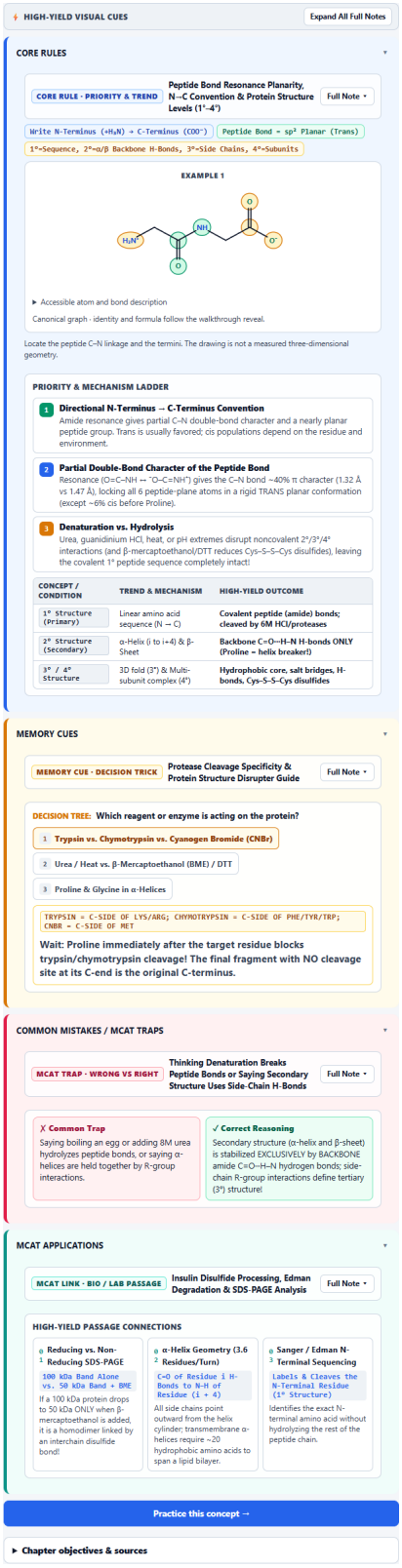
Visual card: Insulin Disulfide Processing, Edman Degradation & SDS-PAGE Analysis

Reducing vs. Non-Reducing SDS-PAGE — 100 kDa Band Alone vs. 50 kDa Band + BME — If a 100 kDa protein drops to 50 kDa ONLY when β-mercaptoethanol is added, it is a homodimer linked by an interchain disulfide bond!

α-Helix Geometry (3.6 Residues/Turn) — C=O of Residue i H-Bonds to N–H of Residue (i + 4) — All side chains point outward from the helix cylinder; transmembrane α-helices require ~20 hydrophobic amino acids to span a lipid bilayer.

Sanger / Edman N-Terminal Sequencing — Labels & Cleaves the N-Terminal Residue (1° Structure) — Identifies the exact N-terminal amino acid without hydrolyzing the rest of the peptide chain.

Screenshot: peptides-visuals



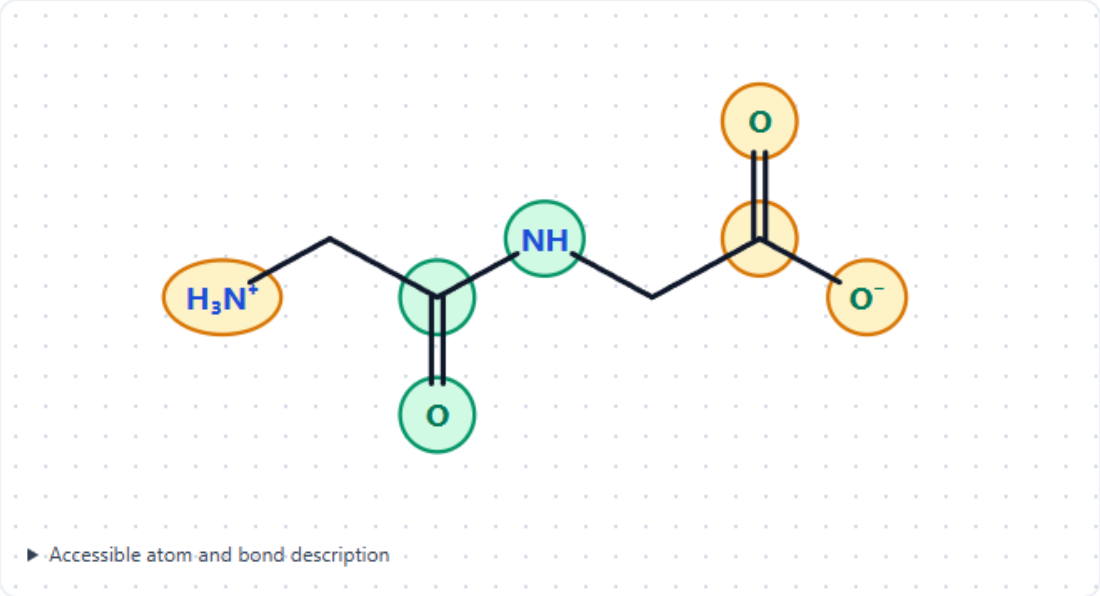

## peptides:step-1 | 1. Find the amide linkage

The internal nitrogen is bonded directly to the preceding carbonyl carbon. The terminal amino and carboxyl groups remain available for ionization.

Displayed molecule ID: glycylglycine

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Screenshot: peptides-step-1



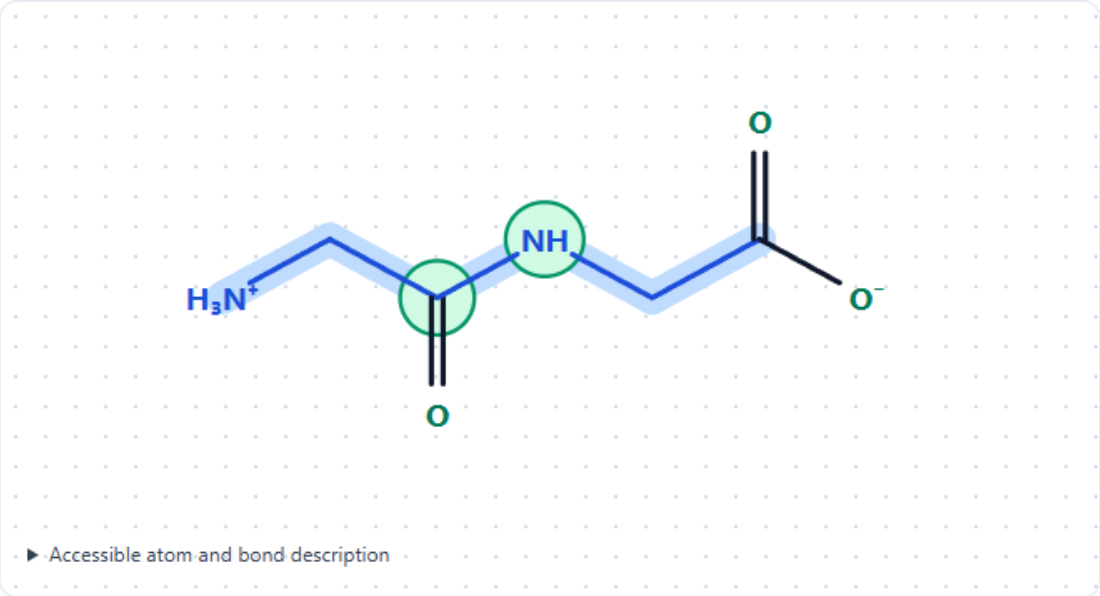

## peptides:step-2 | 2. Trace the backbone

Follow N–Cα–C(=O)–N–Cα–C(=O). The peptide C–N link is approximately planar because the N lone pair participates in resonance.

Displayed molecule ID: glycylglycine

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Screenshot: peptides-step-2



# synthesis | Strecker and Gabriel amino-acid strategies

Track where the carboxyl carbon comes from

## synthesis:rule | Core Rules

In the Strecker route, an aldehyde reacts through an imine/iminium stage with ammonia and cyanide to form an alpha-aminonitrile. Nitrile hydrolysis gives the carboxyl group. A Gabriel–malonic ester route installs protected nitrogen and an alkyl side chain, then hydrolyzes and decarboxylates to an alpha amino acid. One Strecker teaching reagent set uses NH4Cl/KCN, followed by aqueous acid and heat: imine/iminium formation, cyanide carbon addition, then nitrile hydrolysis. These are reaction-recognition details, not an experimental procedure.

Visual card: Strecker & Amidomalonate / Gabriel Amino Acid Syntheses (Preventing Polyalkylation)

Chemistry diagram: Nitrile-carbon tracking through hydrolysis. Product is drawn as a zwitterion after appropriate workup; not a balanced equation.

Strecker: Aldehyde + $NH_3$ + HCN → α-Aminonitrile → (±)-Amino Acid

Gabriel: Phthalimide + Br-Malonate → SN2 + Hydrolysis + ($-CO_2$)

Direct $NH_3$ + R–Br | Uncontrolled polyalkylation! | Gives messy mixture of 1°, 2°, 3° amines + 4° ammonium salt

Strecker Synthesis | 1. RCHO + $NH_3$ + KCN → 2. $H_3O^+$, Δ | Imine + $CN^-$ → α-aminonitrile → hydrolyzed to racemic (±)-amino acid

Gabriel + Malonic Ester | $K^+$Phthalimide + diethyl α-bromomalonate | SN2 + alkylation + $H_3O^+$/Δ ($-CO_2$) → clean 1° (±)-amino acid

Why Direct Alkylation ($NH_3$ + RX) Fails for 1° Amines — — Once $RNH_2$ forms, its inductive alkyl group makes it MORE nucleophilic than $NH_3$, causing rapid over-alkylation to $R_2NH$, $R_3N$, and $R_4N^+X^-$.

Strecker Synthesis (Aldehyde + $NH_3$ + HCN, then $H_3O^+$/Δ) — — Step 1: Aldehyde + $NH_3$ ⇌ Imine (RCH=NH). Step 2: $CN^-$ attacks imine → α-aminonitrile (RCH($NH_2$)CN). Step 3: Acid hydrolysis converts –CN to –COOH!

Gabriel Phthalimide + Malonic Ester Synthesis — — Potassium phthalimide acts as a bulky, masked $NH_2^-$ nucleophile that can only undergo ONE SN2 reaction on α-bromomalonic ester, followed by α-alkylation (R–X), ester/imide hydrolysis, and thermal decarboxylation ($-CO_2$).

## synthesis:cue | Memory Cues

Strecker: nitrile carbon becomes carboxyl carbon. Gabriel–malonate: alkylation builds the side chain; decarboxylation removes one carboxyl group. In the Gabriel–malonic ester variant, potassium phthalimide substitutes at the brominated carbon of diethyl bromomalonate. Deprotonation of the protected aminomalonate then enables carbon alkylation with a suitable alkyl halide by SN2. Hydrolysis/deprotection and decarboxylation give the amino acid; the phthalimide framework can leave as phthalic acid under acidic hydrolysis conditions. The electrophile must be compatible with SN2; tertiary halides do not serve as a universal choice.

Visual card: Tracing Every Atom in Strecker vs. Gabriel-Malonate Amino Acid Synthesis

Where does each piece of the synthesized α-amino acid come from?

Strecker Synthesis: Aldehyde R–CHO + $NH_3$ + KCN — R–CH comes from Aldehyde; $-NH_3^+$ from $NH_3$; –COOH from Cyanide ($CN^-$)! — Wait: The aldehyde carbonyl carbon becomes the α-CARBON (C2) of the amino acid, while the cyanide carbon becomes the C1 CARBOXYLIC ACID!

Gabriel Malonic Ester Synthesis + R–Br — Glycine Backbone from Bromomalonate; Side Chain R from R–Br! — Deprotonating the α-carbon of N-phthalimidomalonic ester with NaOEt followed by SN2 with R–Br installs the side chain R, and $H_3O^+$/Δ cleaves the esters + decarboxylates one –COOH.

What Is the Stereochemistry of Uncatalyzed Strecker or Gabriel Products? — Achiral conditions: racemic chiral product; glycine is achiral — Achiral starting materials produce a planar imine (Strecker) or planar enol during decarboxylation (Gabriel), giving 0% ee unless resolved!

## synthesis:trap | Common Mistakes / MCAT Traps

Achiral starting materials and conditions usually give a racemic chiral amino-acid product; glycine is achiral. Ordinary Gabriel amine synthesis and the malonic-ester amino-acid variant are related but not identical.

Visual card: Thinking the Aldehyde Carbon Becomes the –COOH Group in the Strecker Synthesis

Common Trap — — Assuming the C=O carbon of R–CHO becomes the carboxylic acid in a Strecker synthesis, or forgetting the +1 carbon added by $CN^-$.

Correct Reasoning — — In the Strecker synthesis, the aldehyde carbon becomes the α-carbon (CH–R), and the cyanide (–C≡N) carbon is hydrolyzed into the –COOH group (total carbons = Aldehyde C + 1)!

## synthesis:app | MCAT Applications

Laboratory racemic synthesis contrasts with enzyme-controlled production of stereochemically selected amino acids.

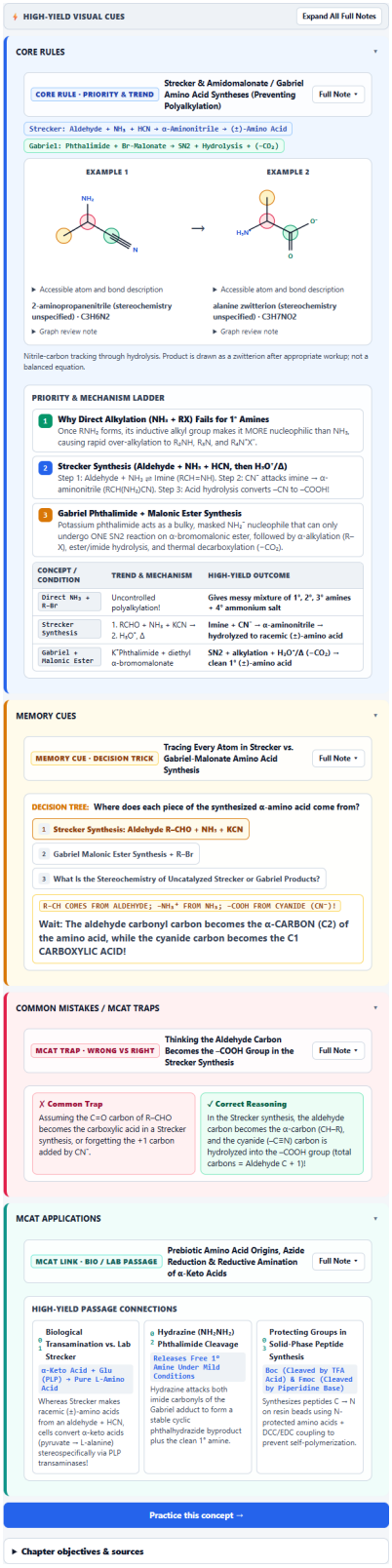
Visual card: Prebiotic Amino Acid Origins, Azide Reduction & Reductive Amination of α-Keto Acids

Biological Transamination vs. Lab Strecker — α-Keto Acid + Glu (PLP) → Pure L-Amino Acid — Whereas Strecker makes racemic (±)-amino acids from an aldehyde + HCN, cells convert α-keto acids (pyruvate → L-alanine) stereospecifically via PLP transaminases!

Hydrazine ($NH_2NH_2$) Phthalimide Cleavage — Releases Free 1° Amine Under Mild Conditions — Hydrazine attacks both imide carbonyls of the Gabriel adduct to form a stable cyclic phthalhydrazide byproduct plus the clean 1° amine.

Protecting Groups in Solid-Phase Peptide Synthesis — Boc (Cleaved by TFA Acid) & Fmoc (Cleaved by Piperidine Base) — Synthesizes peptides C → N on resin beads using N-protected amino acids + DCC/EDC coupling to prevent self-polymerization.

Screenshot: synthesis-visuals




## synthesis:step-1 | 1. Aminonitrile intermediate

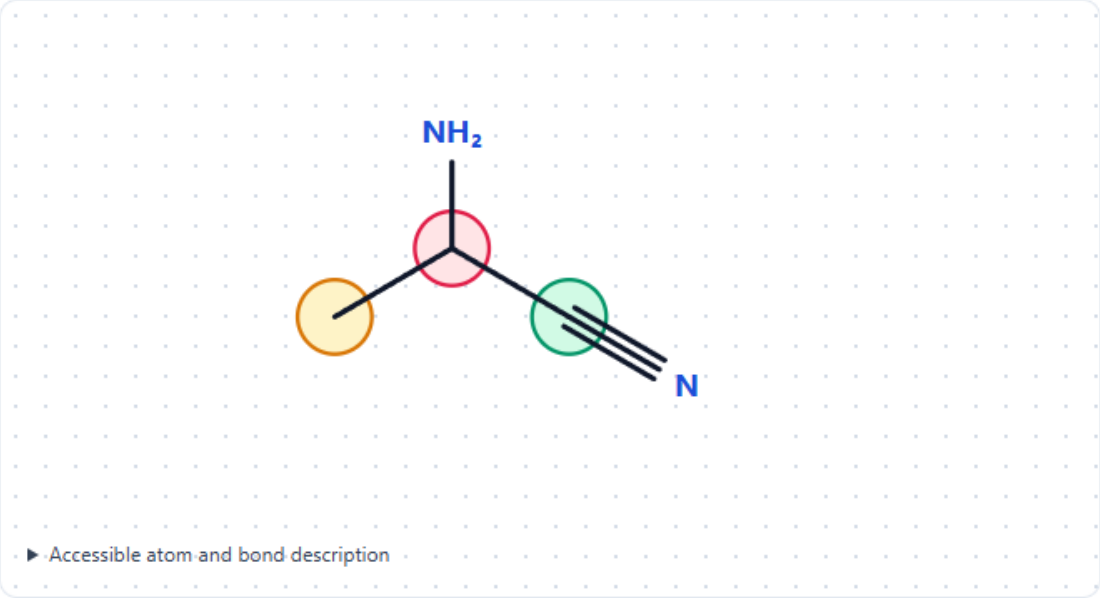
The central carbon carries an amino group, methyl, H, and a nitrile carbon. This is the carbon skeleton leading to alanine on nitrile hydrolysis. Green tracks the nitrile-derived future carboxyl carbon, red the future alpha carbon, and orange the

retained methyl side chain. The nitrile N is not retained in the amino acid.

Displayed molecule ID: aminonitrile

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Screenshot: synthesis-step-1



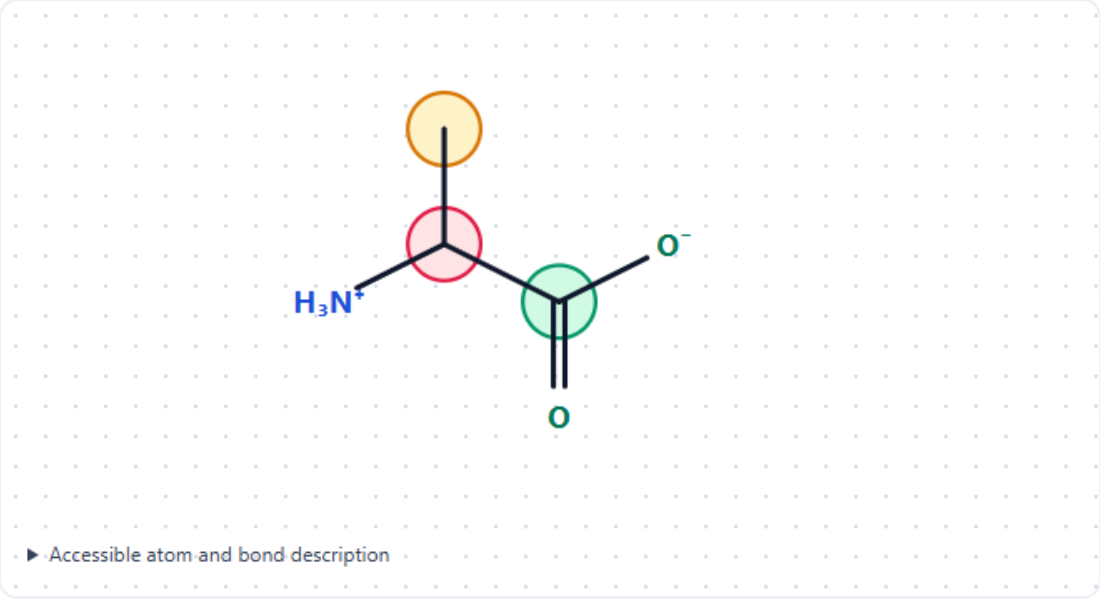

## synthesis:step-2 | 2. Carbon bookkeeping

The nitrile carbon becomes the amino acid's carboxyl carbon. The original aldehyde carbon becomes its alpha carbon; the methyl side chain is retained. Protonation state then depends on pH.

Displayed molecule ID: alanine

{"highlights": {"prin": [2], "sub": [5], "alpha": [1]}, "revealName": true}

Screenshot: synthesis-step-2



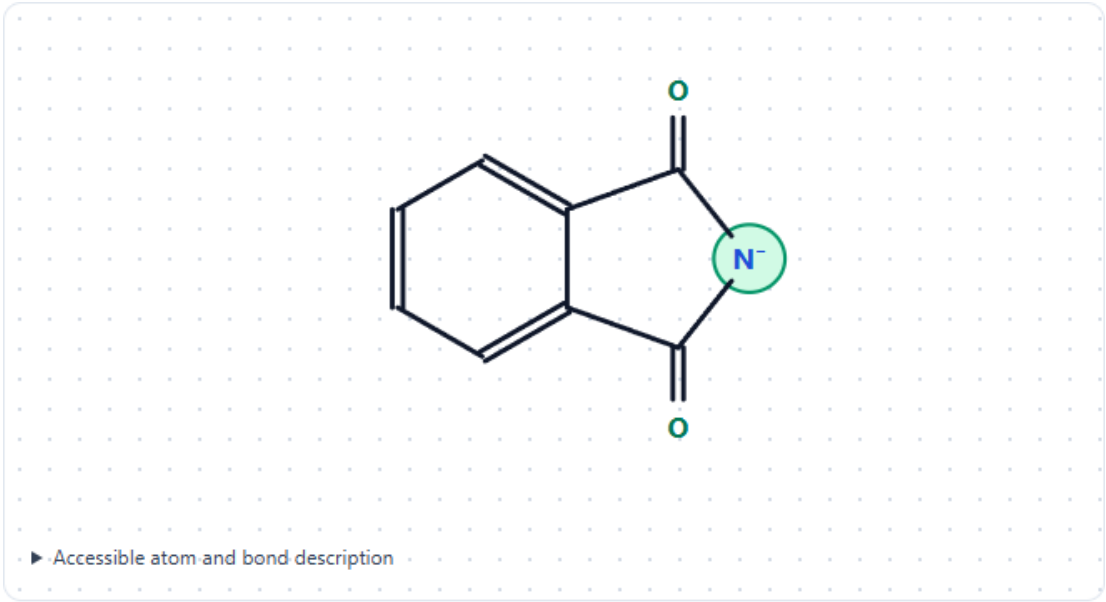

## synthesis:step-3 | 3. Alternative protected route

The phthalimide anion shown supplies protected nitrogen; its potassium counterion is omitted. N substitution on diethyl bromomalonate installs the protected amino group. Subsequent malonate carbon alkylation installs a side chain; hydrolysis/deprotection and decarboxylation then give an alpha amino acid. This reagent canvas is not the complete reaction sequence.

Displayed molecule ID: phthalimide-anion

{"highlights": {"prin": [7]}, "revealName": true}

Screenshot: synthesis-step-3



# phosphorus | Phosphates and phosphoryl transfer

Phosphoesters differ from phosphoanhydrides

## phosphorus:rule | Core Rules

A phosphate ester contains P–O–C; a phosphodiester has two such ester links. A phosphoanhydride contains P–O–P. Phosphoric acid has successive dissociations, so charge depends on pH. Formal P=O Lewis drawings are bookkeeping models of delocalized bonding. Organic phosphate esters attach carbon groups through P–O–C. Inorganic orthophosphate (Pi) lacks such ester links and includes protonation states set by pH. Pyrophosphate (PPi) joins two phosphate units through P–O–P; its charge also depends on pH.

Visual card: Phosphoric Acid Speciation (pKa 2.1, 7.2, 12.4), ATP Bioenergetics & Phosphodiester Stability

Chemistry diagram: Phosphoric acid and a phosphate ester: formal Lewis representations, not localized-bonding measurements.

$H_3PO_4$: pKa1 ~2.1, pKa2 ~7.2, pKa3 ~12.4

pH 7.4 Buffer = $H_2PO_4^- \rightleftharpoons HPO_4^{2-}$

ATP = 2 Phosphoanhydride + 1 Phosphoester

pKa1 (~2.1) | $H_3PO_4 \rightleftharpoons H_2PO_4^- + H^+$ | Fully deprotonated at pH 7.4

pKa2 (~7.2) | $H_2PO_4^- \rightleftharpoons HPO_4^{2-} + H^+$ | Ideal physiological buffer at pH 7.4!

pKa3 (~12.4) | $HPO_4^{2-} \rightleftharpoons PO_4^{3-} + H^+$ | Stays protonated at pH 7.4

DNA Phosphodiester | $RO–PO_2^-–OR'$ (−1 charge per link) | Negative charge repels $OH^-$ hydrolysis & keeps DNA inside cell

Three Separate Buffering Zones of Phosphoric Acid ($H_3PO_4$) — — Phosphoric acid is triprotic with widely spaced pKa values (~2.1, ~7.2, ~12.4); it buffers ONLY within ±1 pH unit of each individual pKa (NOT uniformly across pH 2–12!). At pH 7.4, $H_2PO_4^-$ and $HPO_4^{2-}$ predominate.

Why ATP Hydrolysis Is Highly Exergonic ($\Delta G°' \approx -30.5$ kJ/mol) — — 1) Relieves electrostatic repulsion among 4 clustered negative charges on α/β/γ phosphates; 2) Inorganic phosphate (P_i) and ADP have superior resonance stabilization (less competing oxygen sharing); 3) Greater solvation & entropy.

Kinetic Stability of DNA/RNA Phosphodiester Backbones — — At pH 7.4, every phosphodiester linkage carries a −1 charge ($RO–PO_2^-–OR'$), which electrostatically repels nucleophilic attack by water/hydroxide (RNA hydrolyzes faster in base only because its 2′-OH acts as an internal nucleophile!).

## phosphorus:cue | Memory Cues

DNA/RNA backbone: phosphodiester. Between ATP phosphate groups: phosphoanhydride. Approximate aqueous phosphoric-acid pKa values are 2.1, 7.2 and 12.3; $H_2PO_4^-/HPO_4^{2-}$ buffers near neutral pH. Temperature and ionic conditions shift the values. Useful buffering regions lie near each pKa, often within about one pH unit; capacity depends on concentration and component ratio. Three separated regions do not give uniformly effective buffering at all intervening pH values.

Visual card: ATP α / β / γ Phosphate Attack & RNA vs. DNA Alkaline Stability

What phosphoryl transfer or nucleic acid question is asked?

Kinase Phosphoryl Transfer vs. Synthetase Adenylation — Kinase Attacks γ-P (Releases ADP); Adenylation Attacks α-P (Releases PP_i → 2 P_i)! — Attacking α-P releases pyrophosphate (PP_i), which inorganic pyrophosphatase immediately hydrolyzes to 2 P_i—providing a double thermodynamic boost!

Why Does RNA Hydrolyze in Base (NaOH) While DNA Survives? — RNA Has a 2′-OH Group (DNA Is 2′-Deoxy!) — Base deprotonates the ribose 2′-OH into a 2′-alkoxide right next door, which attacks the phosphorus to form a 2′,3′-cyclic phosphate and cleave the RNA chain!

Pentavalent Trigonal Bipyramidal Transition State (SN2-Like at P) — In-Line Attack by Nucleophile Opposite the Leaving Group (with $Mg^{2+}$) — $Mg^{2+}$ chelates the negatively charged phosphate oxygens, neutralizing charge repulsion and polarizing P=O for nucleophilic attack.

## phosphorus:trap | Common Mistakes / MCAT Traps

Breaking a bond requires energy. Favorable ATP hydrolysis reflects the overall free-energy difference between reactants and solvated products, including resonance, electrostatics, and solvation; energy is not released simply by breaking one bond.

Visual card: Calling ATP Bonds "High-Energy Because Breaking Bonds Releases Energy"

Common Trap — — Saying "breaking the P–O bond in ATP releases energy" or claiming phosphate buffers uniformly from pH 2 to pH 12.

Correct Reasoning — — Breaking ANY chemical bond is endothermic (requires energy)! ATP hydrolysis releases free energy because the products (ADP + P_i) form stronger bonds, have less charge repulsion, and enjoy greater resonance/solvation than ATP!

## phosphorus:app | MCAT Applications

Phosphoryl transfer can couple reactions. Nucleic-acid synthesis releases pyrophosphate, whose subsequent hydrolysis can help favor polymerization.

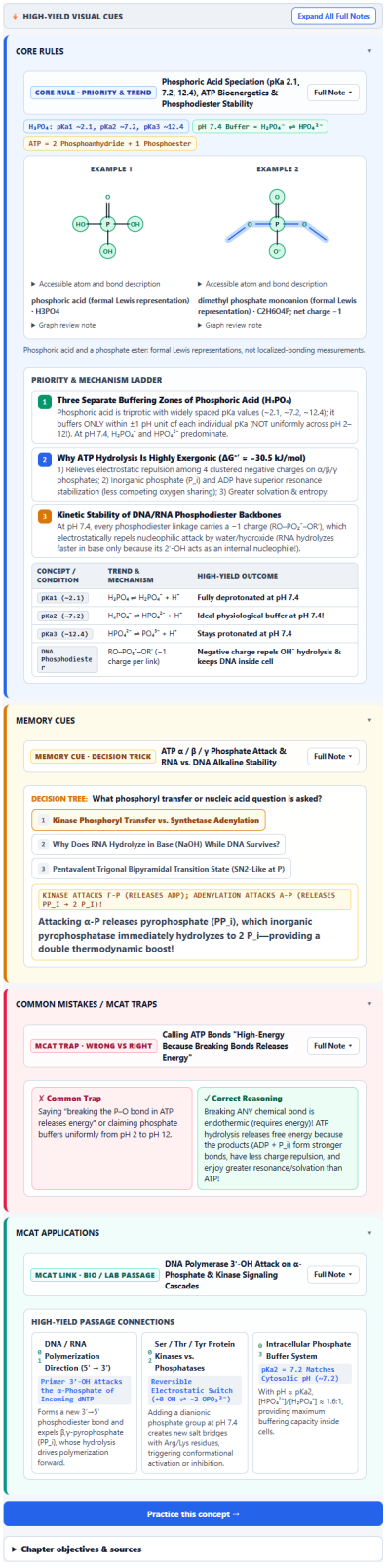
Visual card: DNA Polymerase 3′-OH Attack on α-Phosphate & Kinase Signaling Cascades

DNA / RNA Polymerization Direction (5′ → 3′) — Primer 3′-OH Attacks the α-Phosphate of Incoming dNTP — Forms a new 3′→5′ phosphodiester bond and expels β,γ-pyrophosphate (PP_i), whose hydrolysis drives polymerization forward.

Ser / Thr / Tyr Protein Kinases vs. Phosphatases — Reversible Electrostatic Switch (+0 OH ⇌ −2 $OPO_3^{2-}$) — Adding a dianionic phosphate group at pH 7.4 creates new salt bridges with Arg/Lys residues, triggering conformational activation or inhibition.

Intracellular Phosphate Buffer System — pKa2 = 7.2 Matches Cytosolic pH (~7.2) — With pH ≈ pKa2, $[HPO_4^{2-}]/[H_2PO_4^{-}]$ ≈ 1.6:1, providing maximum buffering capacity inside cells.

Screenshot: phosphorus-visuals



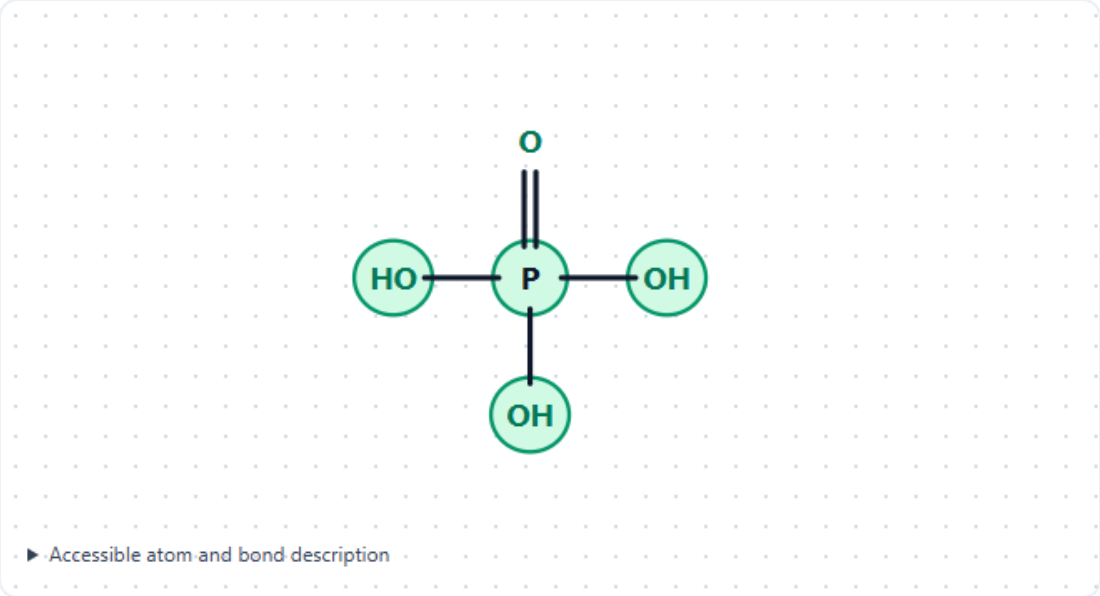

## phosphorus:step-1 | 1. Acidic phosphate

The neutral phosphoric-acid representation has three OH groups. Sequential deprotonation gives different phosphate ions; this drawing is not the predominant form at every pH.

Displayed molecule ID: phosphoricacid

{"highlights": {"prin": [0, 2, 3, 4]}, "revealName": true}

Screenshot: phosphorus-step-1



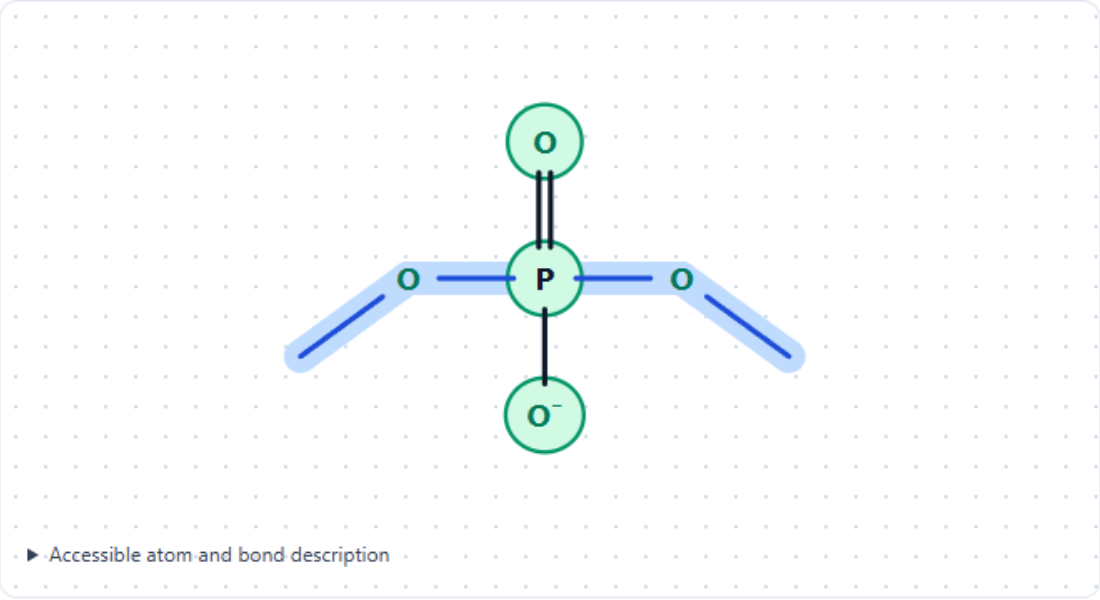

## phosphorus:step-2 | 2. Phosphodiester model

Dimethyl phosphate illustrates two P–O–C connections and a negative charge. The nucleic-acid backbone replaces the methyl groups with sugar-derived groups.

Displayed molecule ID: dimethylphosphate

{"highlights": {"path": [5, 2, 0, 3, 6], "prin": [0, 1, 4]}, "revealName": true}

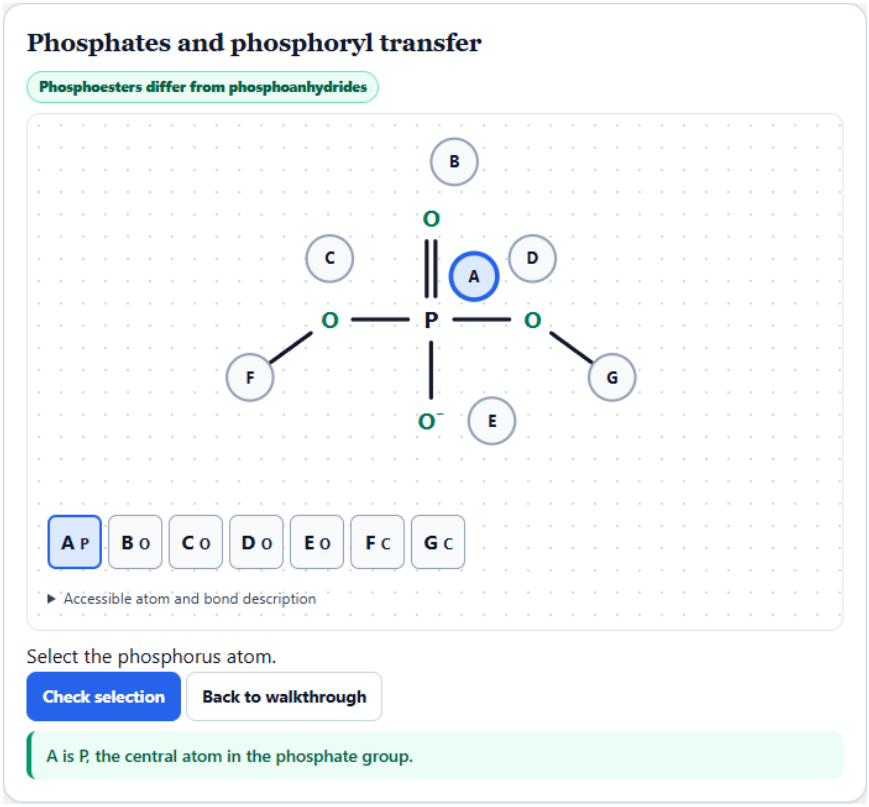
Screenshot: phosphorus-step-2




Embedded challenge: Select the phosphorus atom.

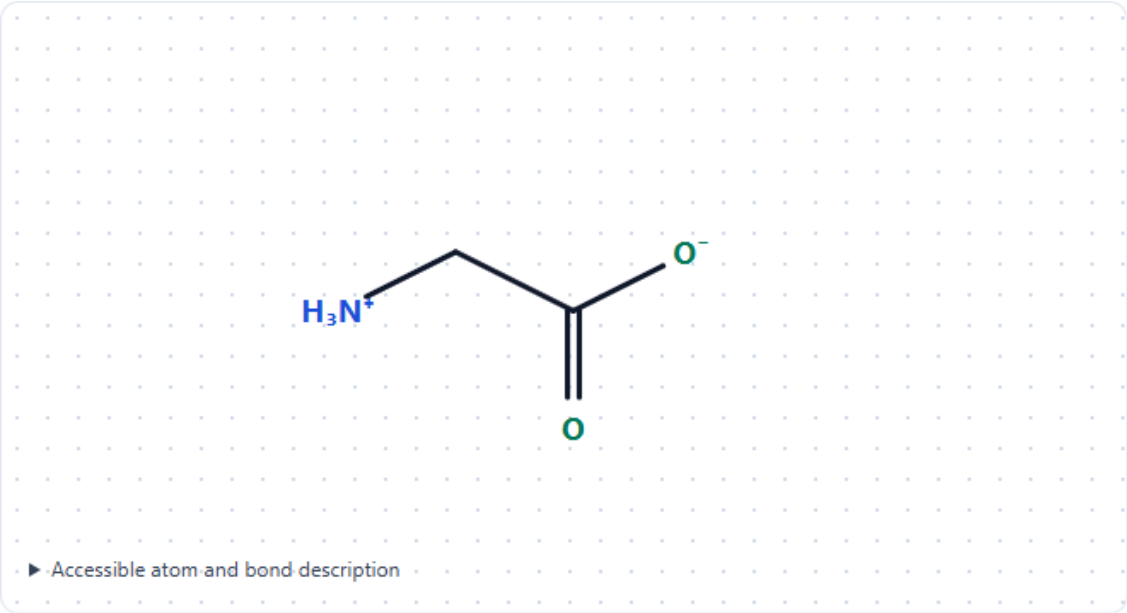
{"prompt": "Select the phosphorus atom.", "answer": [0], "explanation": "A is P, the central atom in the phosphate group."}

Screenshot: phosphorus-step-2-challenge



# glycine-charge | Charge and isoelectric point

Screenshot: glycine-charge-stage-1-molecule




## glycine-charge:stage-1 | What is the net formal charge of the displayed zwitterion?

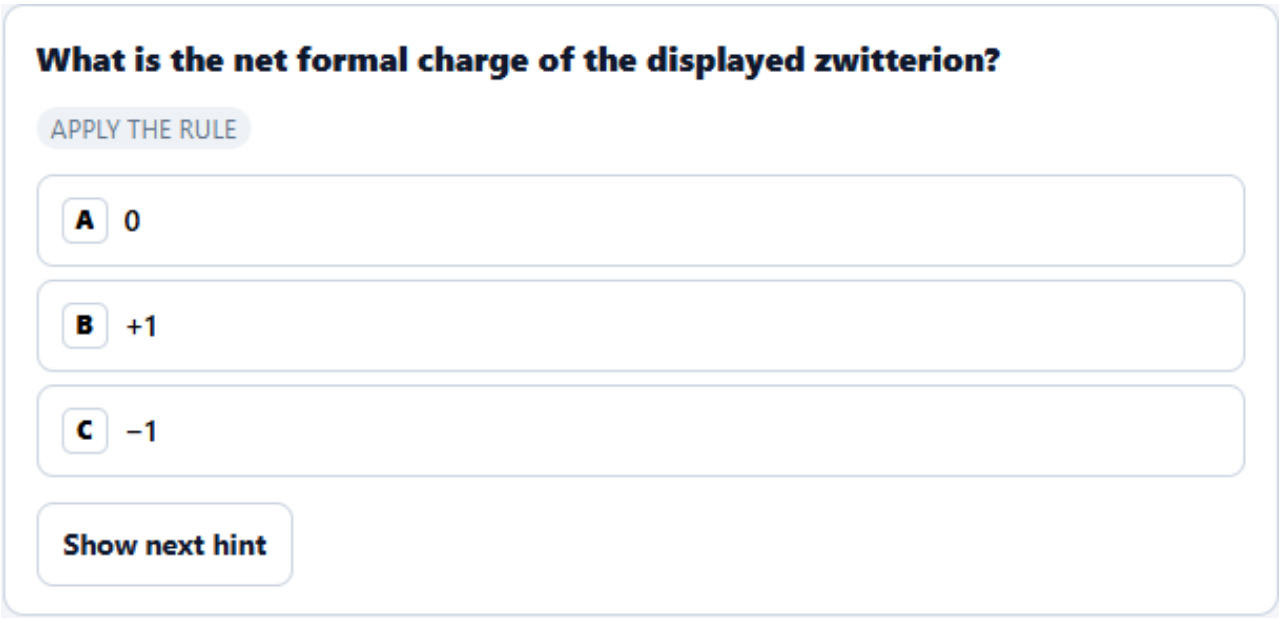
Answer type: choice

Option 0 (A): 0

Option 1 (B): +1

Option 2 (C): −1

Screenshot: glycine-charge-stage-1-question



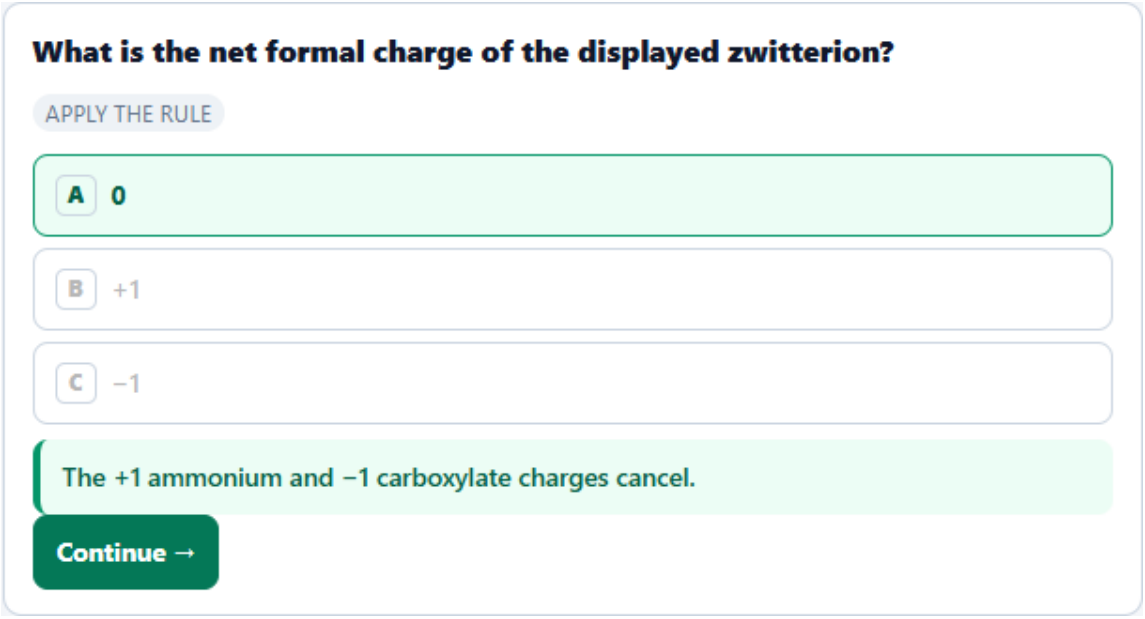

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Add the labeled formal charges.

Hint 2: Net zero does not mean no charged sites.

Explanation: The +1 ammonium and −1 carboxylate charges cancel.

Screenshot: glycine-charge-stage-1-feedback




## glycine-charge:stage-2 | Why is glycine achiral?

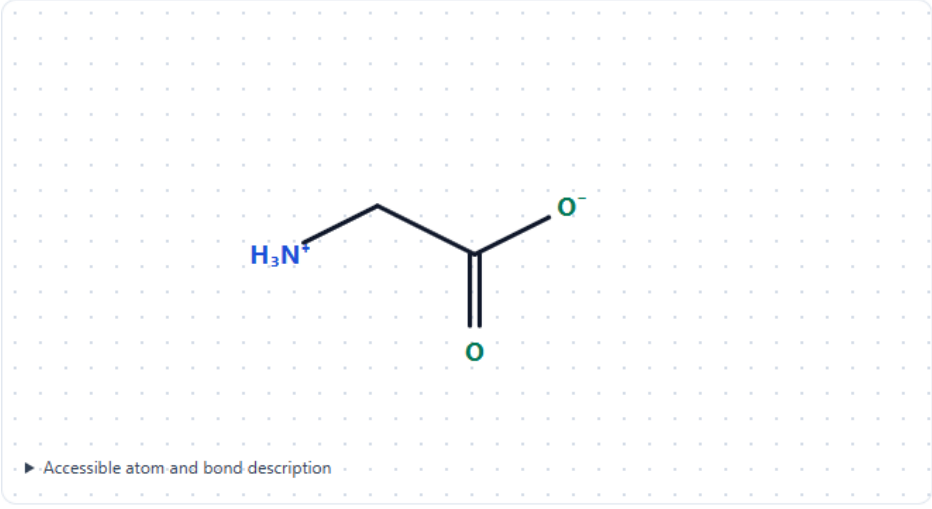
Answer type: choice

Option 0 (A): Its alpha carbon has two H substituents

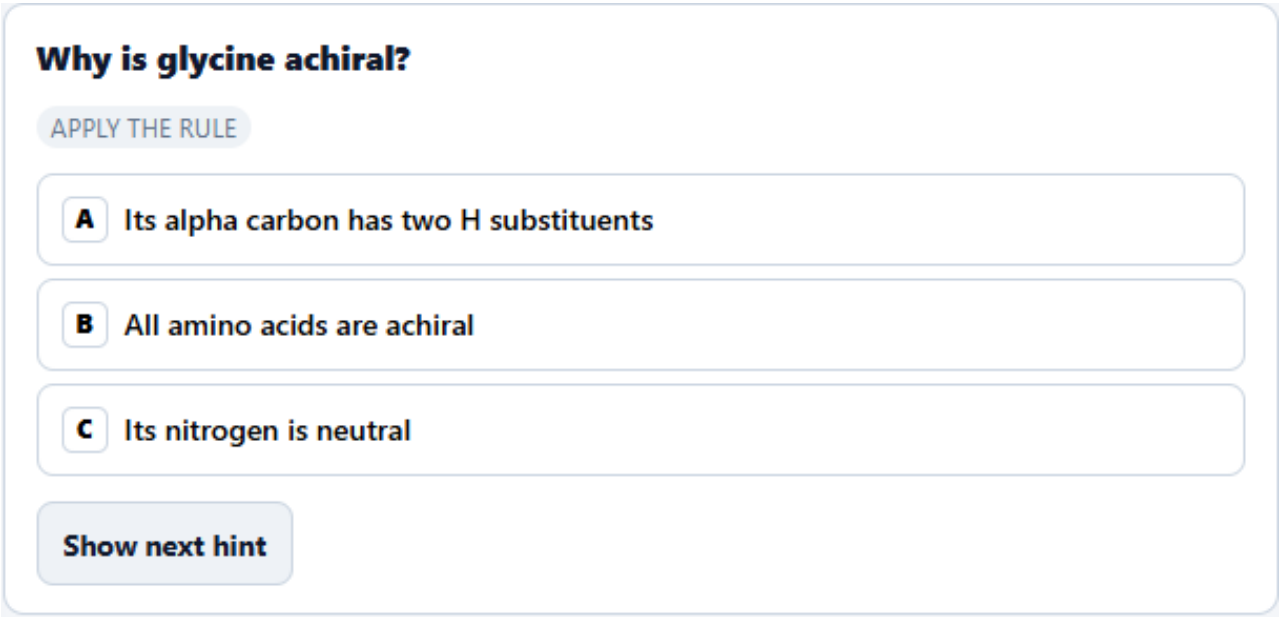
Option 1 (B): All amino acids are achiral

Option 2 (C): Its nitrogen is neutral

Screenshot: glycine-charge-stage-2-molecule



Screenshot: glycine-charge-stage-2-question



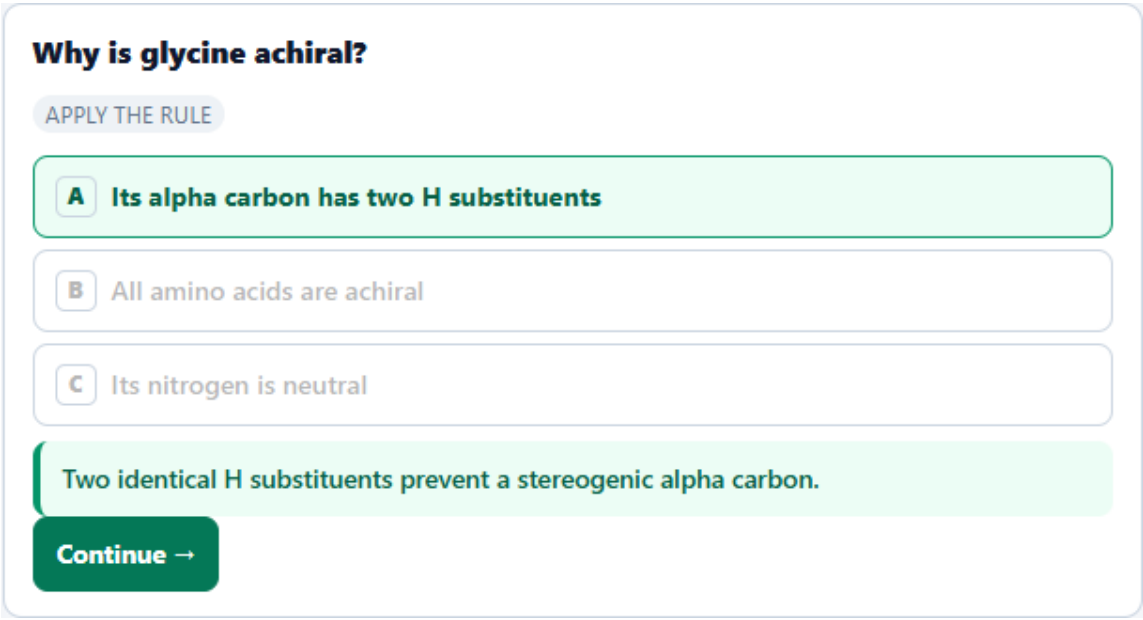

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Inspect the implicit hydrogens.

Hint 2: The side chain itself is H.

Explanation: Two identical H substituents prevent a stereogenic alpha carbon.

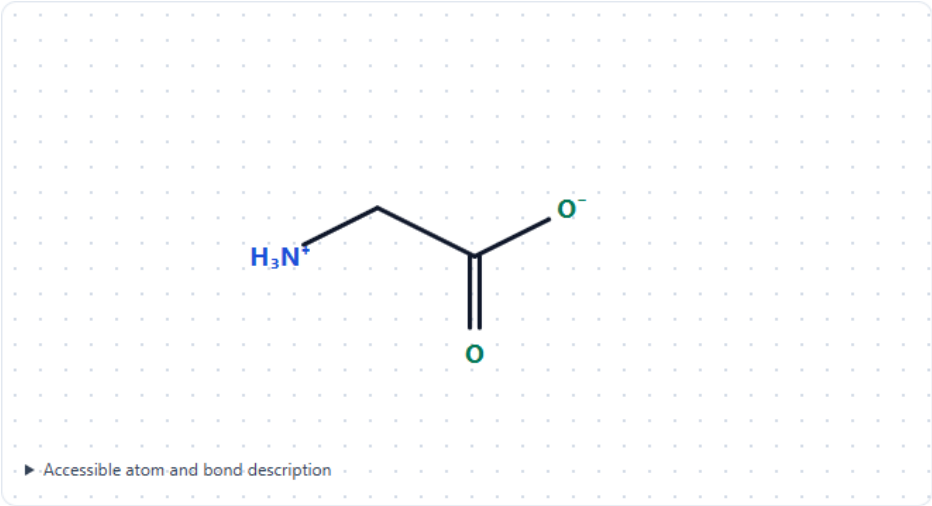
Screenshot: glycine-charge-stage-2-feedback




## glycine-charge:stage-3 | Using model pKa values 2.30 and 9.60, calculate pI. Report two decimal places.

Answer type: number | unit: pH units

Screenshot: glycine-charge-stage-3-molecule




Screenshot: glycine-charge-stage-3-question

**Using model pKa values 2.30 and 9.60, calculate pI. Report two decimal places.**

CALCULATE AND INTERPRET

Numeric answer (pH units)

Check answer

Show next hint

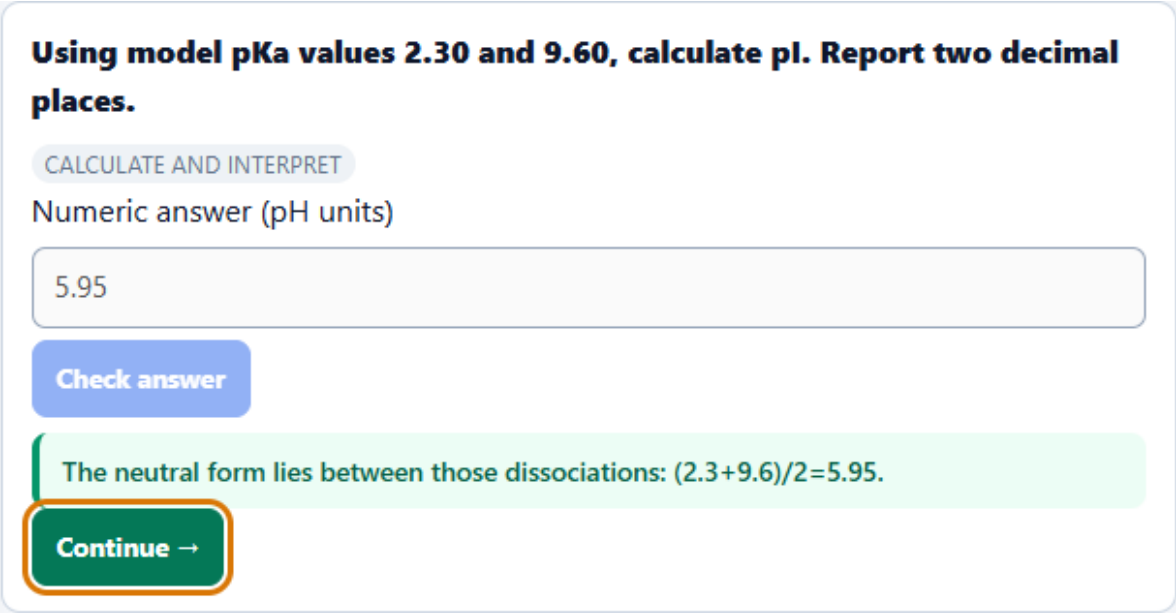
{"authorAnswer": 5.95, "tolerance": 0.005, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the pKa pair bracketing net zero.

Hint 2: Average the two supplied numbers; retain two decimal places.

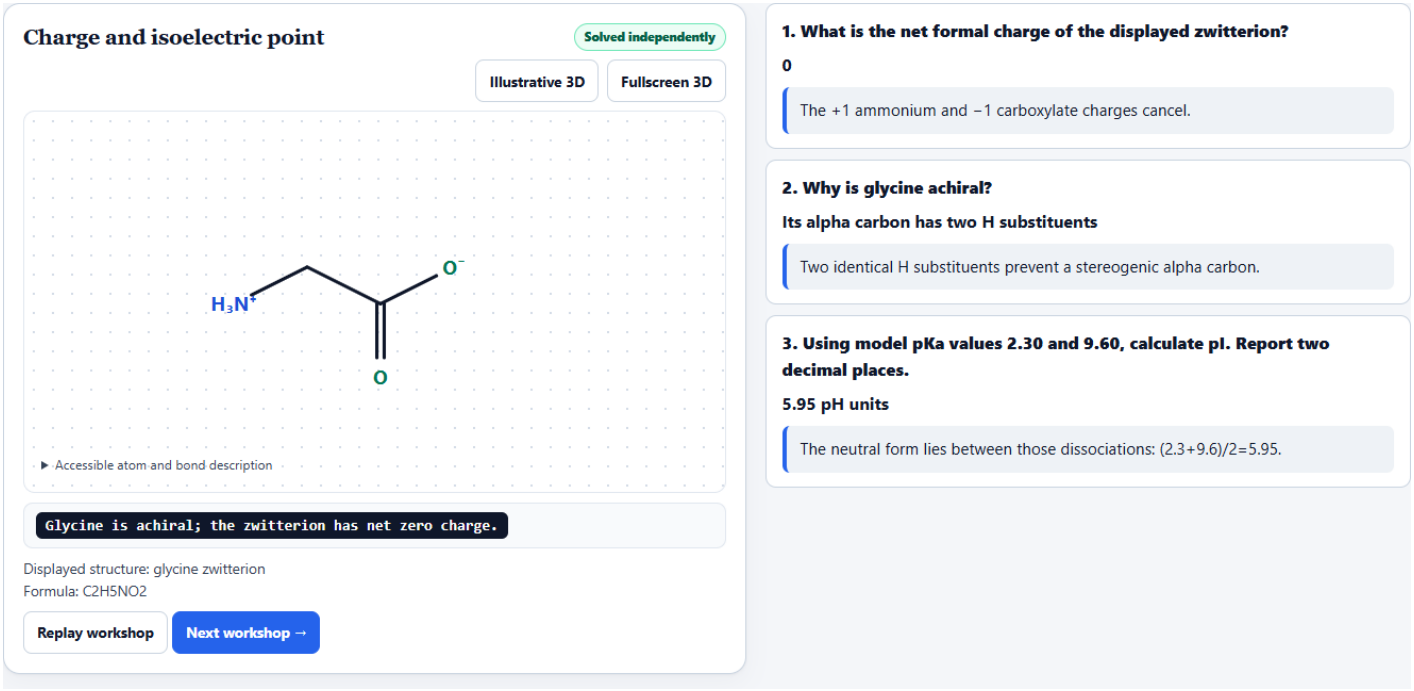
Explanation: The neutral form lies between those dissociations: (2.3+9.6)/2=5.95.

Screenshot: glycine-charge-stage-3-feedback



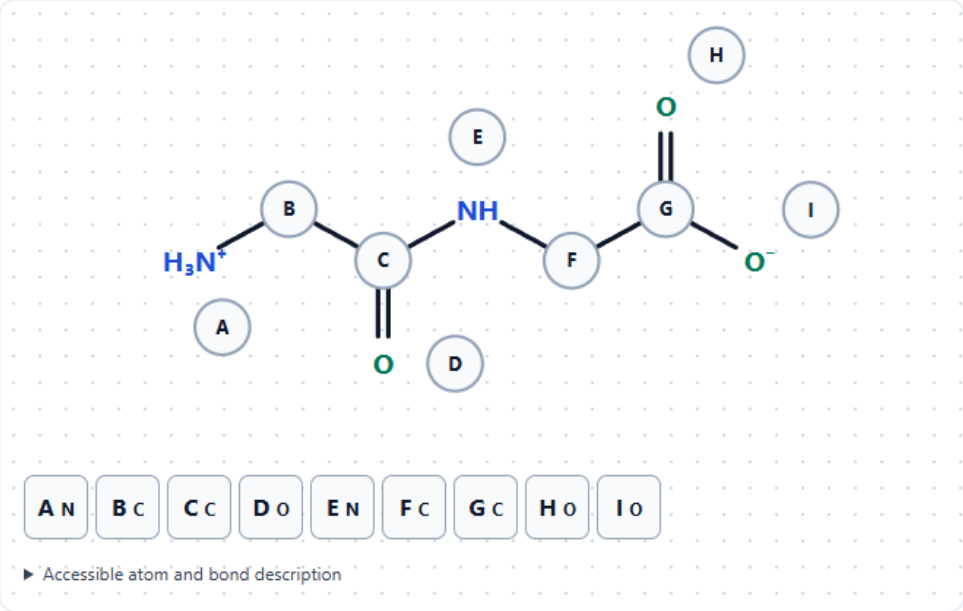

Completion summary: Glycine is achiral; the zwitterion has net zero charge.

Screenshot: glycine-charge-complete



# peptide-link | Read a peptide backbone

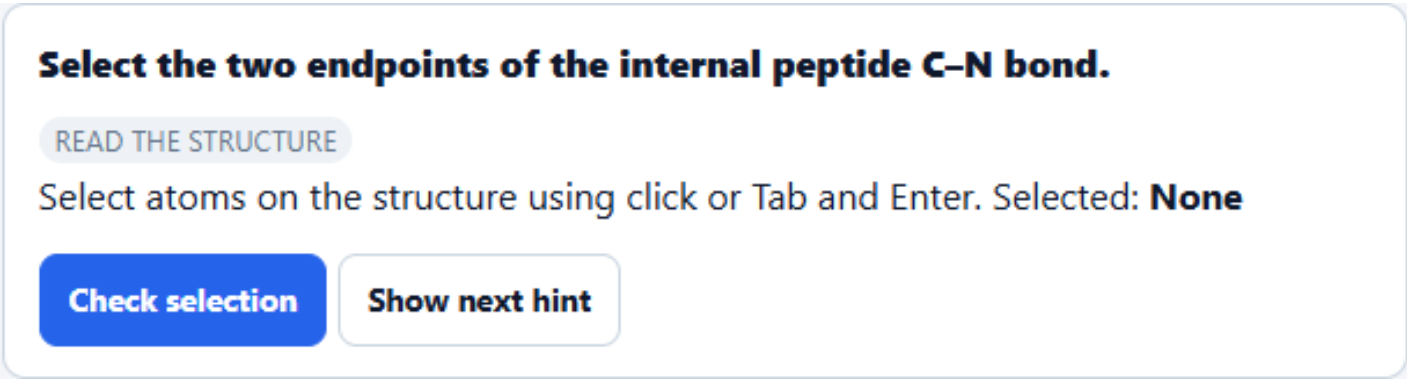
Screenshot: peptide-link-stage-1-molecule




## peptide-link:stage-1 | Select the two endpoints of the internal peptide C–N bond.

Answer type: atoms

Screenshot: peptide-link-stage-1-question



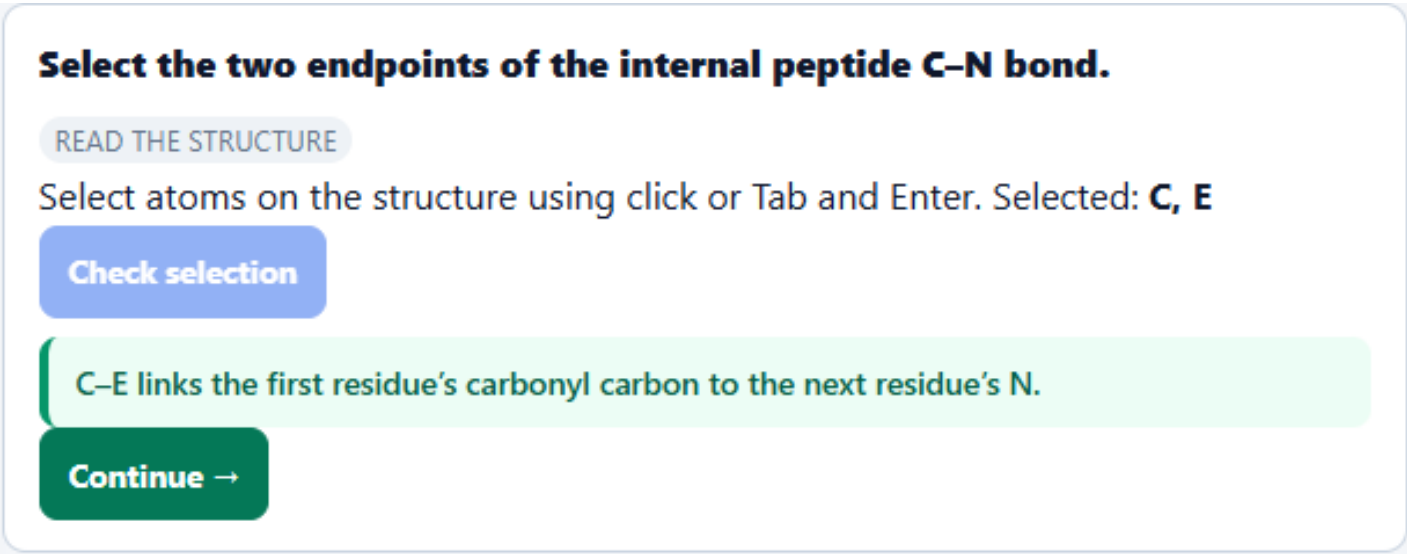

{"authorAnswer": [2, 4], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [2, 4]}}

Hint 1: Find C(=O)–N inside the molecule.

Hint 2: Exclude the terminal ammonium nitrogen.

Explanation: C–E links the first residue's carbonyl carbon to the next residue's N.

Screenshot: peptide-link-stage-1-feedback




## peptide-link:stage-2 | Why is rotation around this bond restricted?

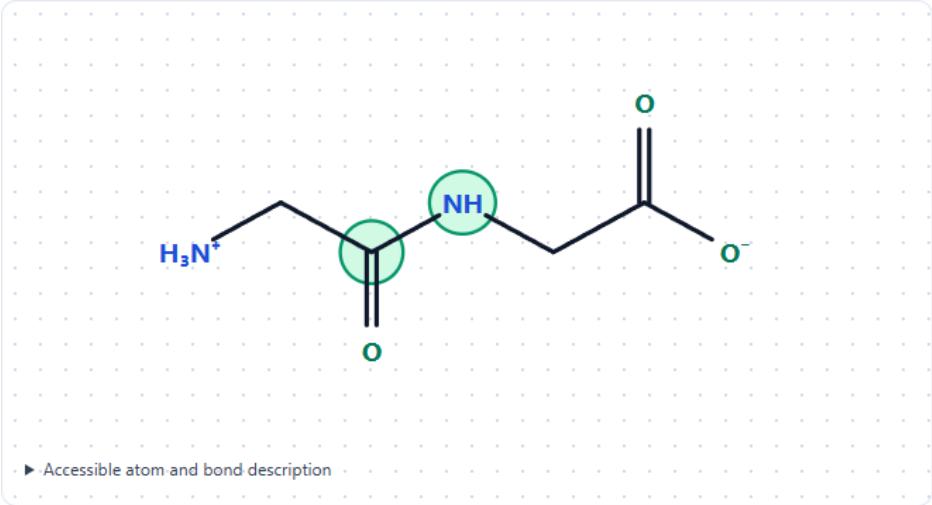
Answer type: choice

Option 0 (A): Amide resonance gives partial double-bond character

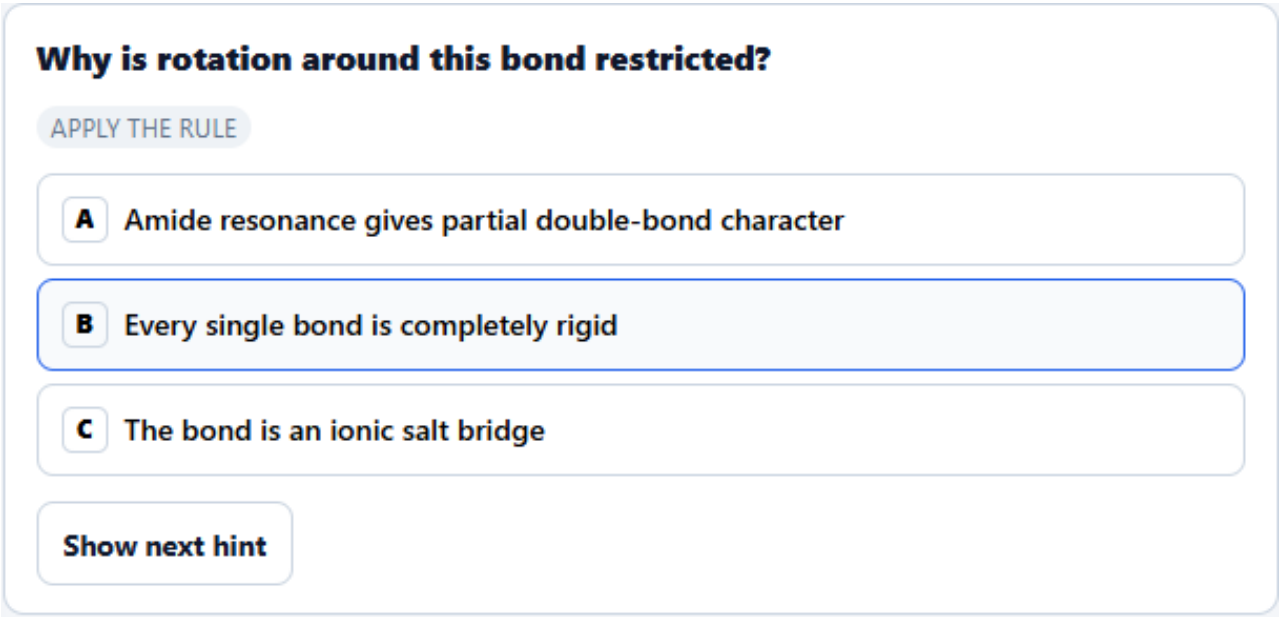
Option 1 (B): Every single bond is completely rigid

Option 2 (C): The bond is an ionic salt bridge

Screenshot: peptide-link-stage-2-molecule



Screenshot: peptide-link-stage-2-question



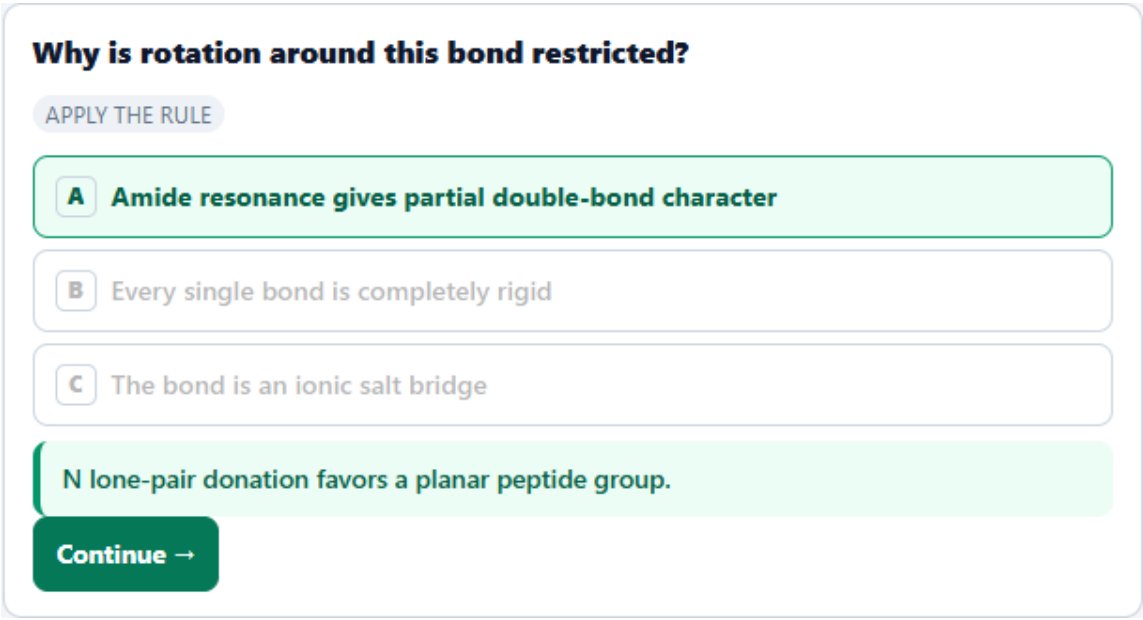

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Apply the amide resonance model.

Hint 2: Aligned p orbitals stabilize the group.

Explanation: N lone-pair donation favors a planar peptide group.

Screenshot: peptide-link-stage-2-feedback



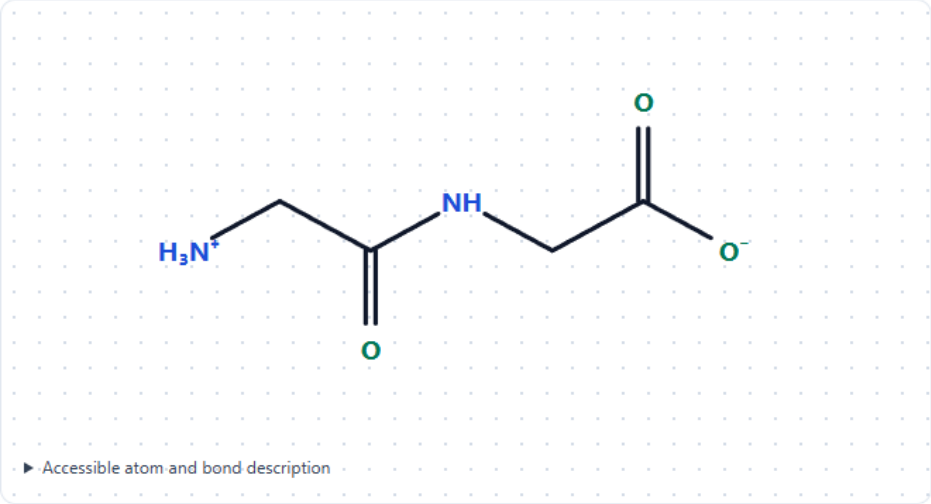

## peptide-link:stage-3 | Peptide sequences are read from...

Answer type: choice

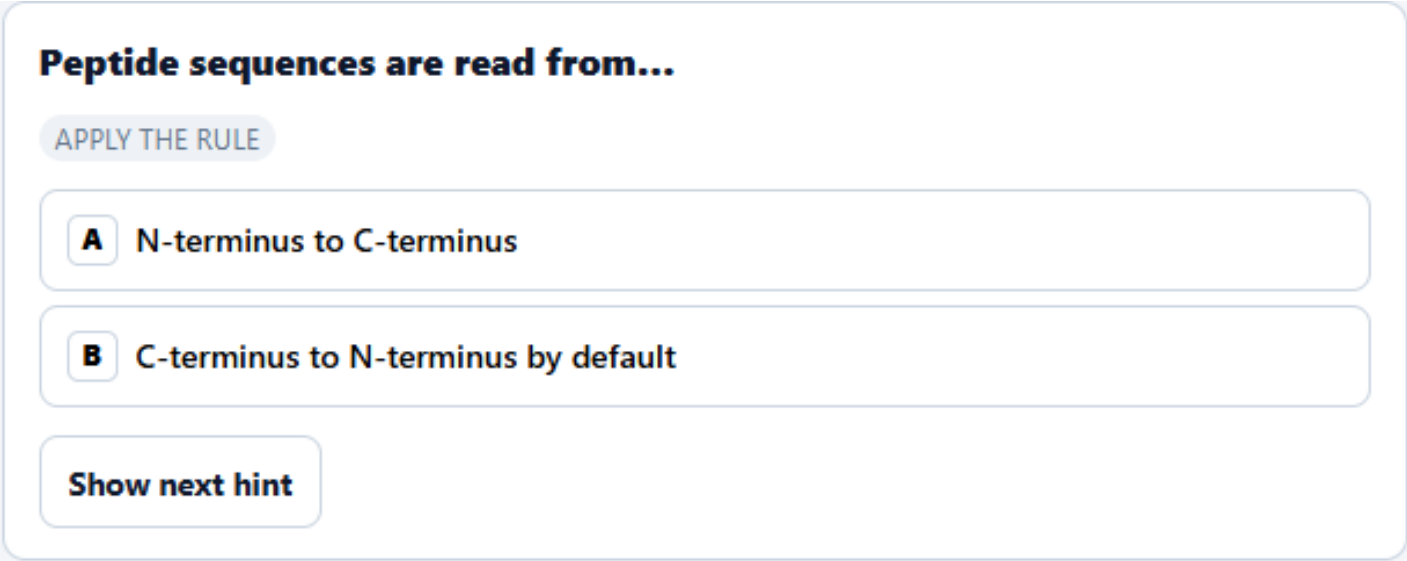
Option 0 (A): N-terminus to C-terminus

Option 1 (B): C-terminus to N-terminus by default

Screenshot: peptide-link-stage-3-molecule




Screenshot: peptide-link-stage-3-question



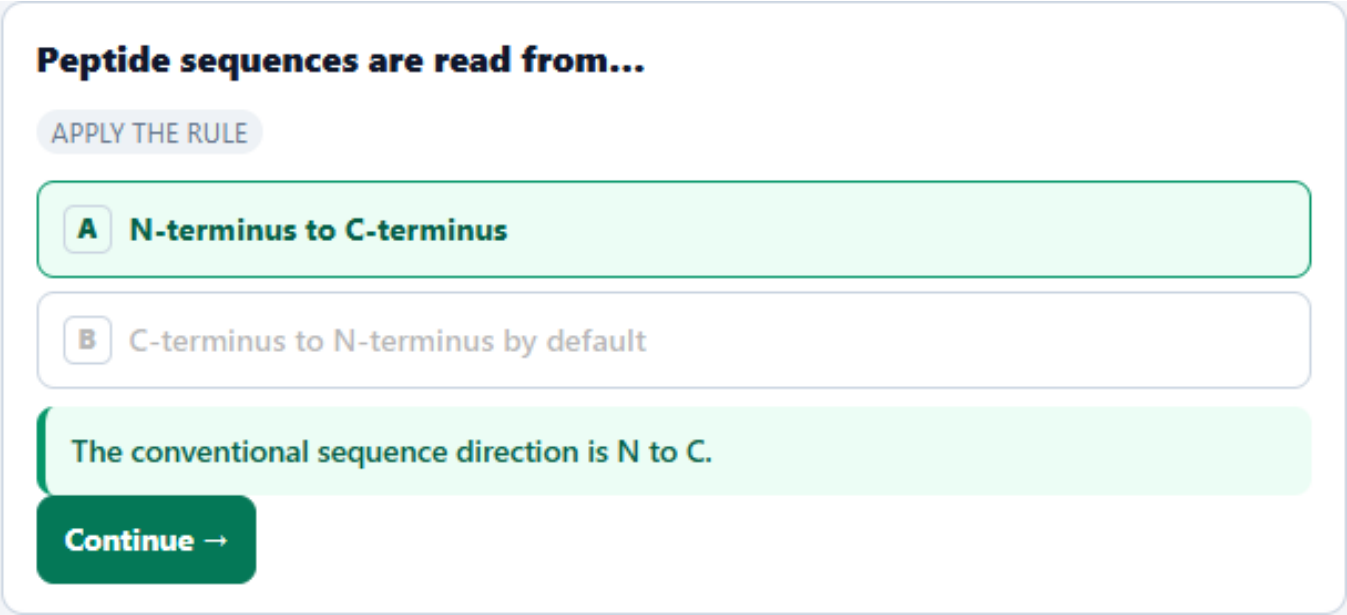

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Find the free terminal amino group.

Hint 2: Start there and follow the backbone.

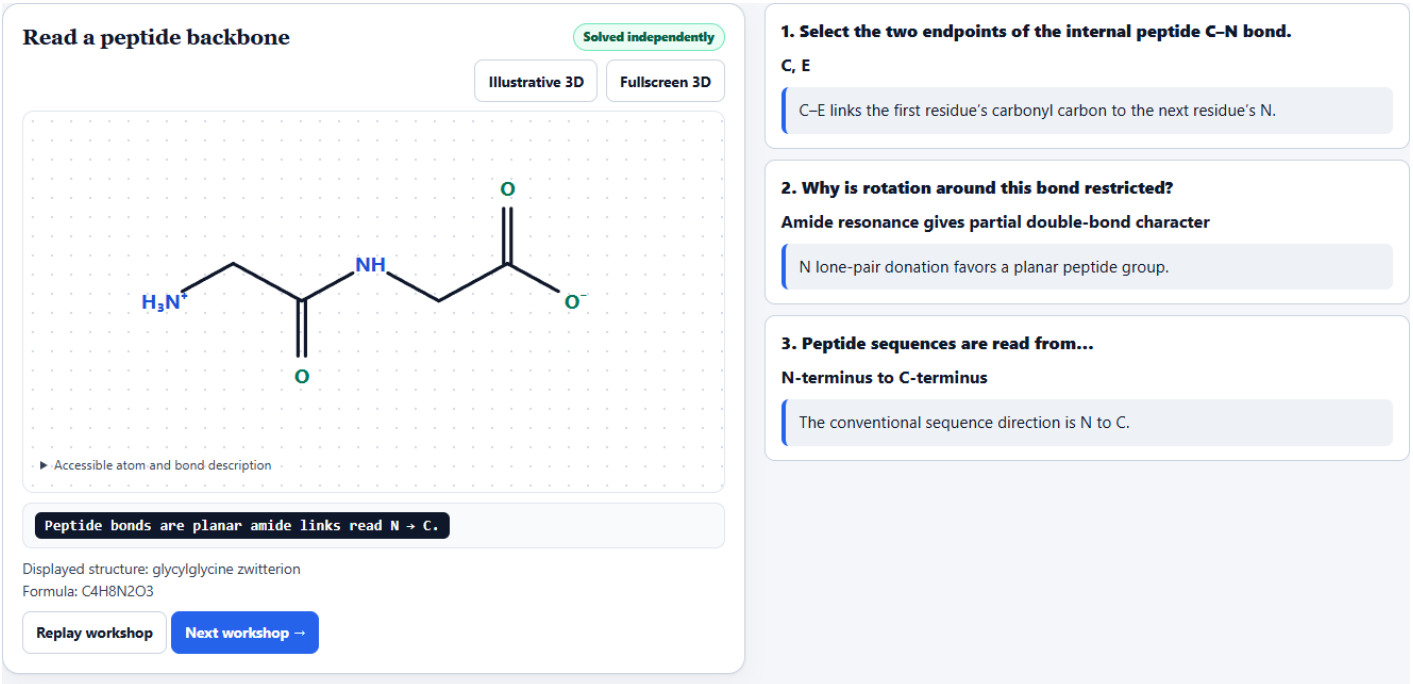
Explanation: The conventional sequence direction is N to C.

Screenshot: peptide-link-stage-3-feedback



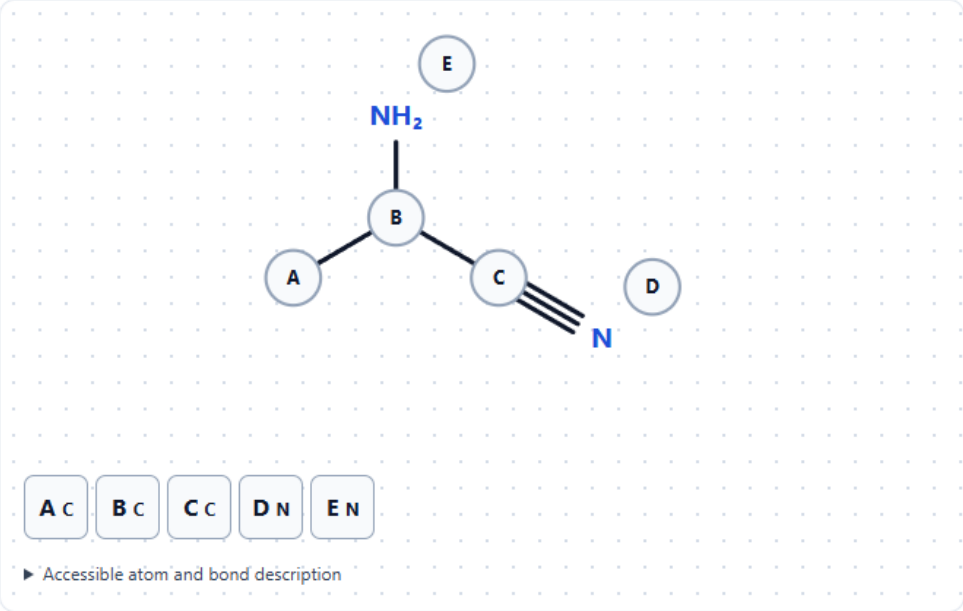

Completion summary: Peptide bonds are planar amide links read N → C.

Screenshot: peptide-link-complete



# strecker-map | Track a Strecker intermediate

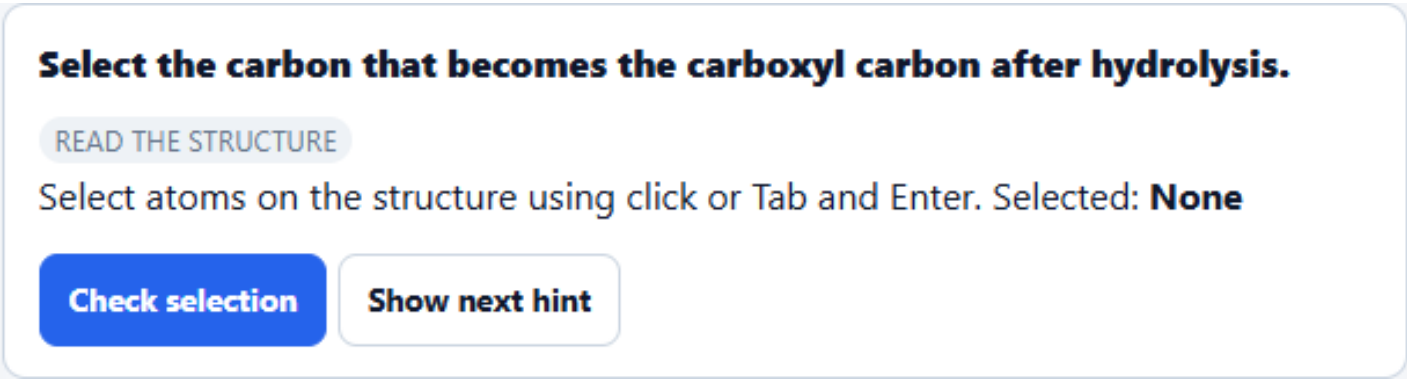
Screenshot: strecker-map-stage-1-molecule




## strecker-map:stage-1 | Select the carbon that becomes the carboxyl carbon after hydrolysis.

Answer type: atoms

Screenshot: strecker-map-stage-1-question



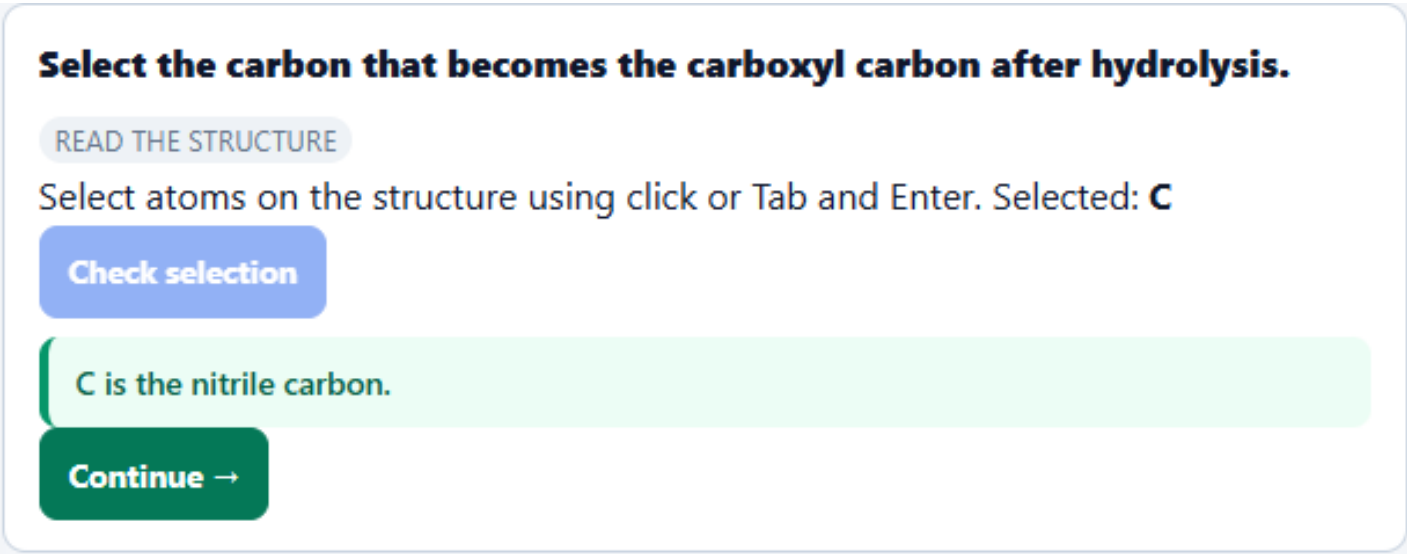

{"authorAnswer": [2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [2]}}

Hint 1: Hydrolysis transforms C≡N into a carboxyl group.

Hint 2: Track the carbon, not the nitrile nitrogen.

Explanation: C is the nitrile carbon.

Screenshot: strecker-map-stage-1-feedback




## strecker-map:stage-2 | Which group is retained as the alanine side chain?

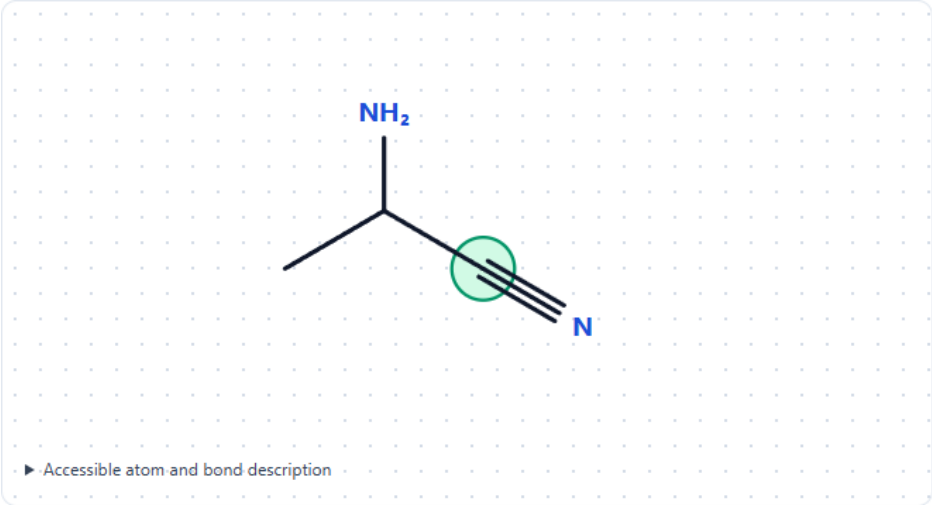
Answer type: choice

Option 0 (A): Methyl

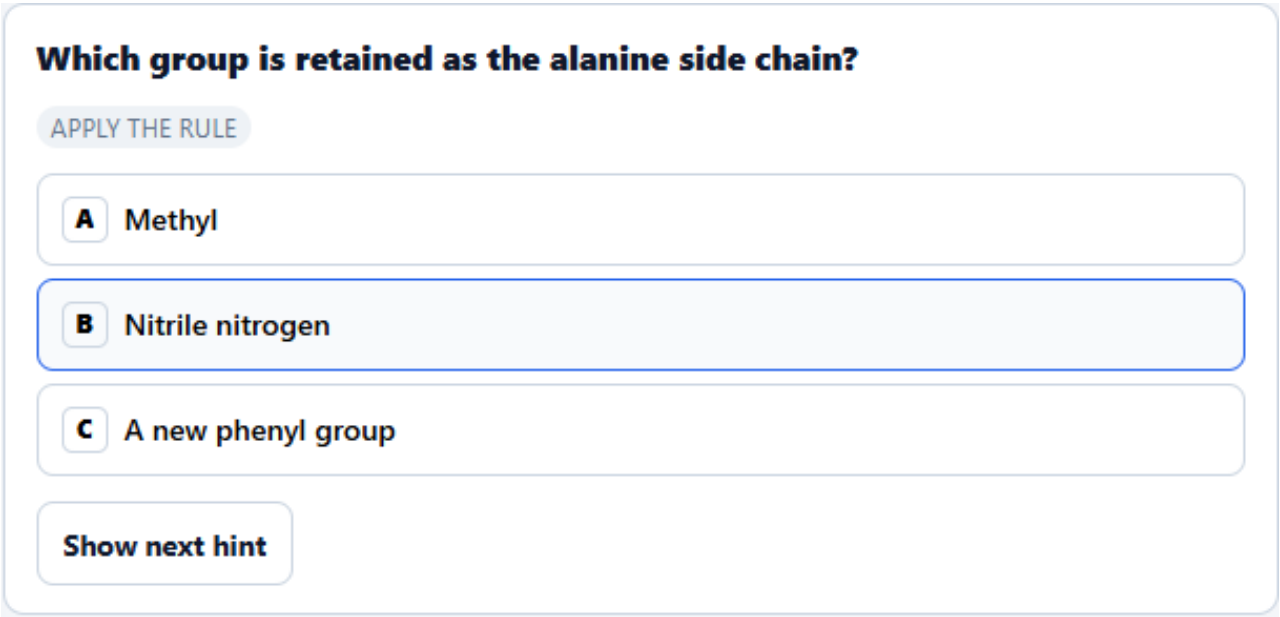
Option 1 (B): Nitrile nitrogen

Option 2 (C): A new phenyl group

Screenshot: strecker-map-stage-2-molecule



Screenshot: strecker-map-stage-2-question



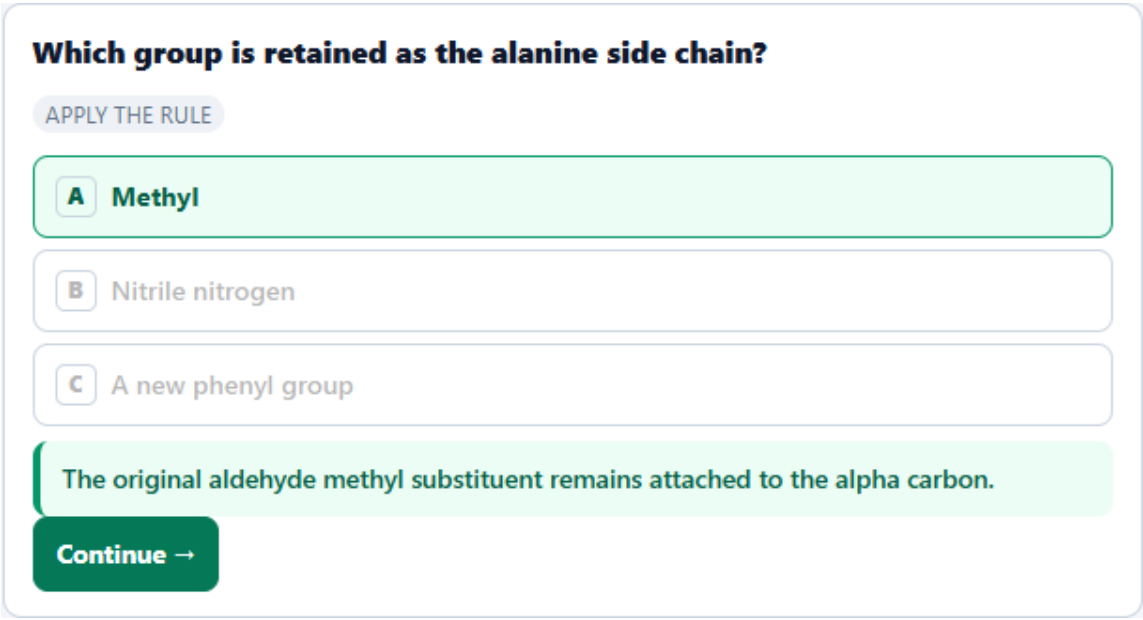

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {"sub": [0]}}

Hint 1: Follow the carbon skeleton.

Hint 2: No new aromatic group is introduced.

Explanation: The original aldehyde methyl substituent remains attached to the alpha carbon.

Screenshot: strecker-map-stage-2-feedback




## strecker-map:stage-3 | Under achiral conditions, the chiral product is generally...

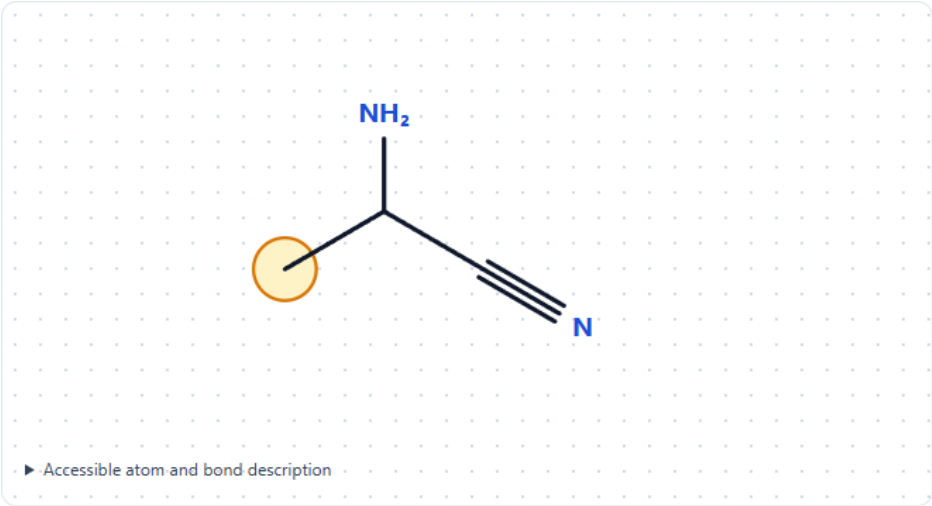
Answer type: choice

Option 0 (A): Racemic

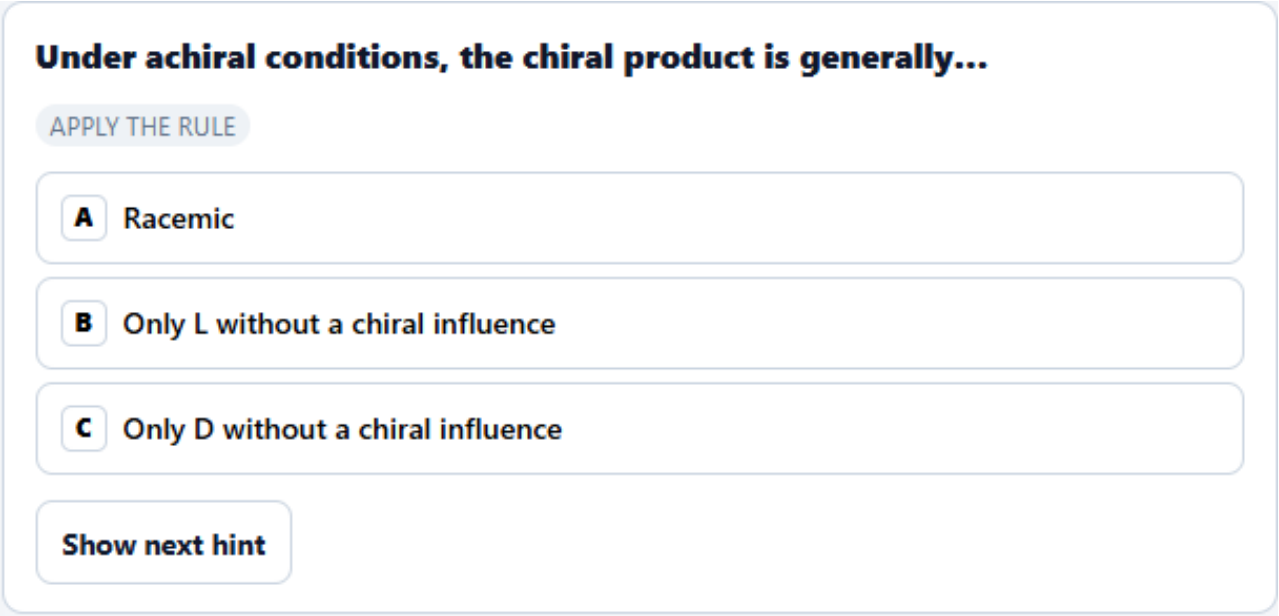
Option 1 (B): Only L without a chiral influence

Option 2 (C): Only D without a chiral influence

Screenshot: strecker-map-stage-3-molecule




Screenshot: strecker-map-stage-3-question



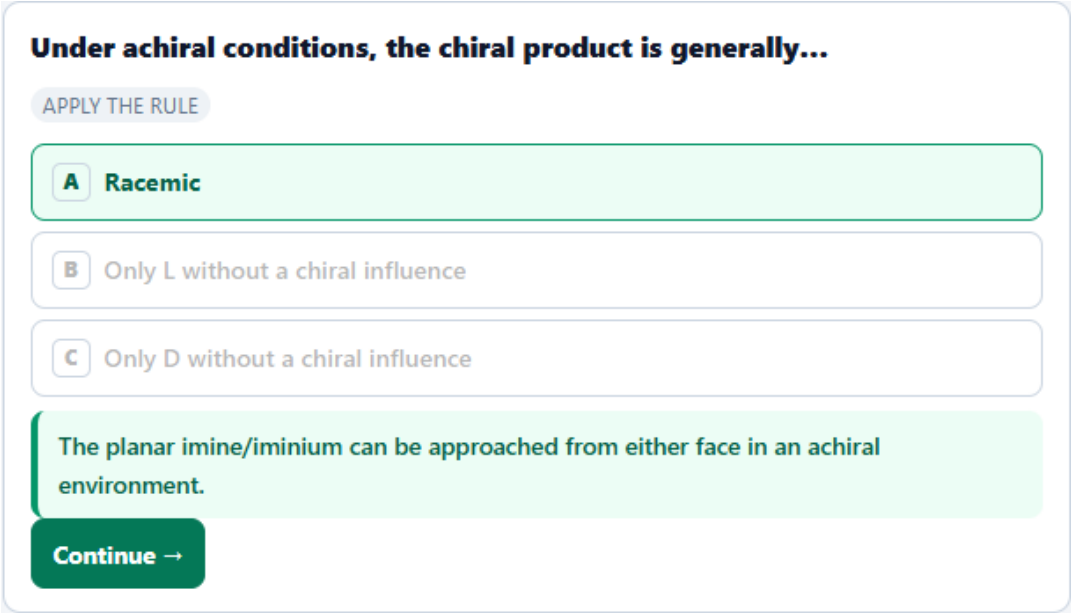

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Ask whether the reaction distinguishes enantiotopic faces.

Hint 2: No chiral catalyst or reagent is specified.

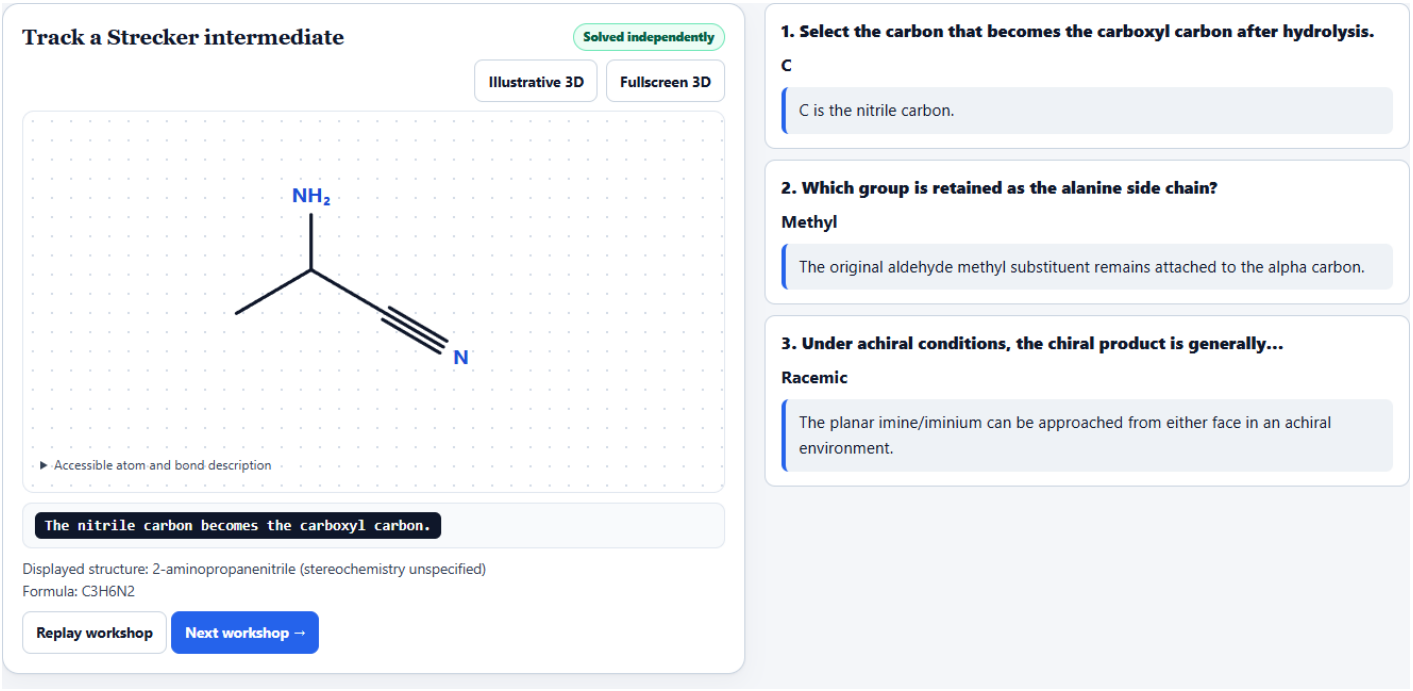
Explanation: The planar imine/iminium can be approached from either face in an achiral environment.

Screenshot: strecker-map-stage-3-feedback



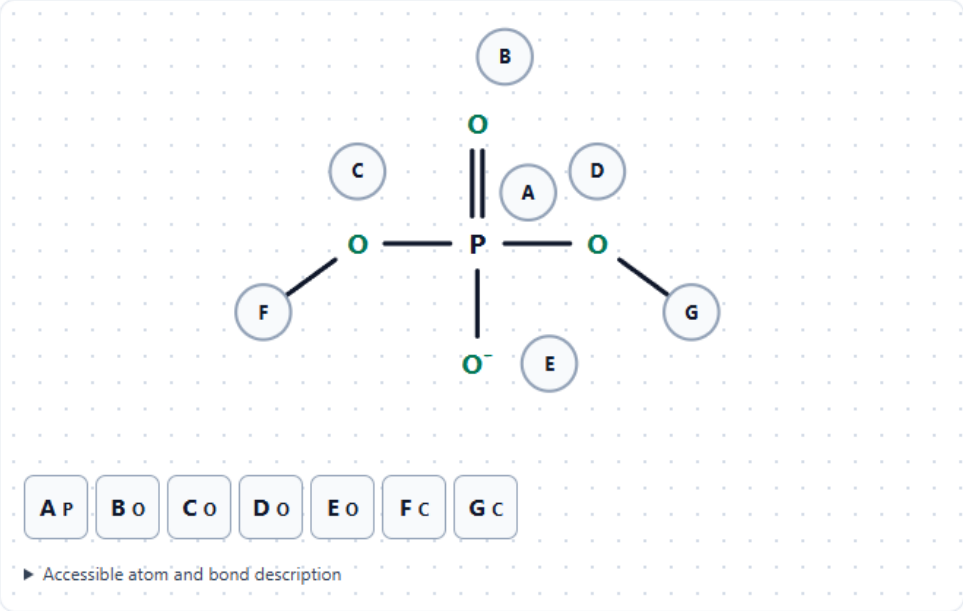

Completion summary: The nitrile carbon becomes the carboxyl carbon.

Screenshot: strecker-map-complete



# phosphate-links | Identify phosphorus linkages

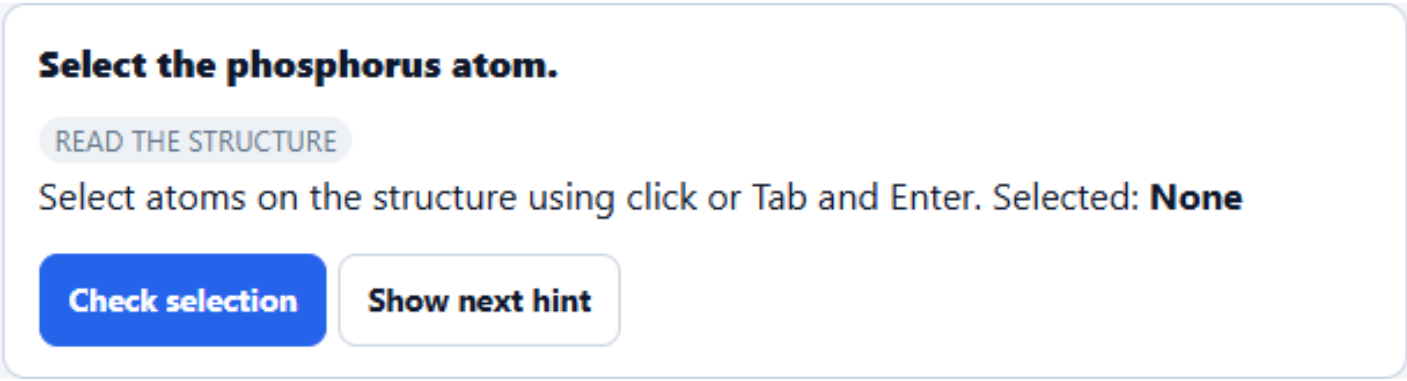
Screenshot: phosphate-links-stage-1-molecule




## phosphate-links:stage-1 | Select the phosphorus atom.

Answer type: atoms

Screenshot: phosphate-links-stage-1-question



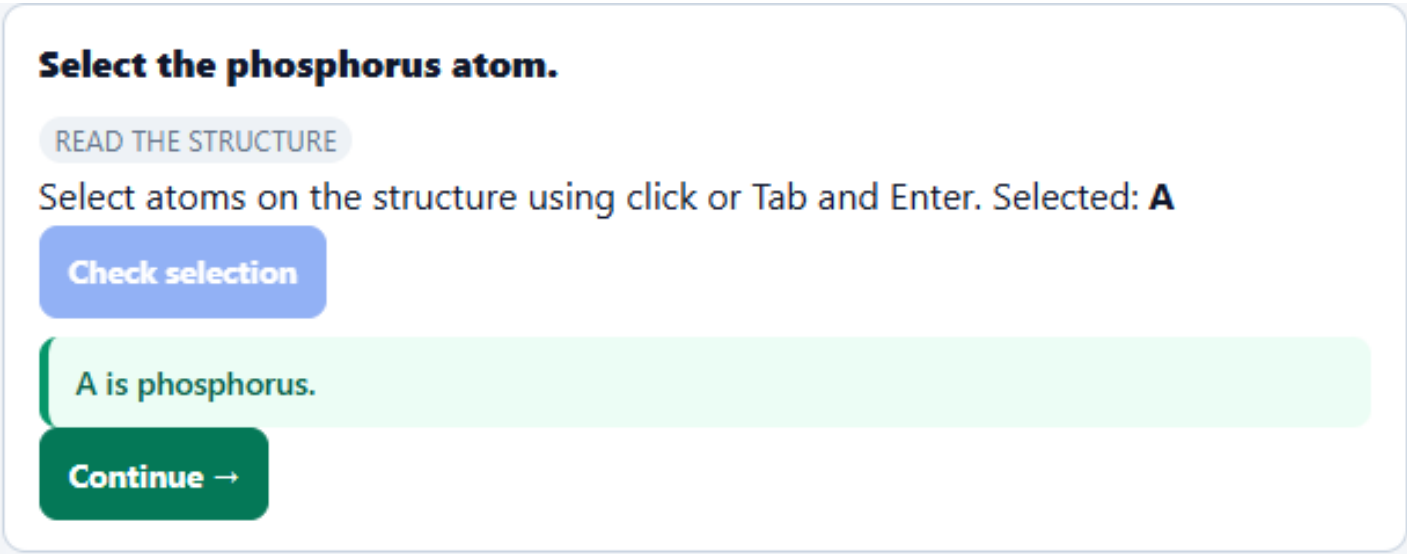

{"authorAnswer": [0], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0]}}

Hint 1: Read the P label.

Hint 2: It is bonded to four oxygens.

Explanation: A is phosphorus.

Screenshot: phosphate-links-stage-1-feedback




## phosphate-links:stage-2 | This structure models a...

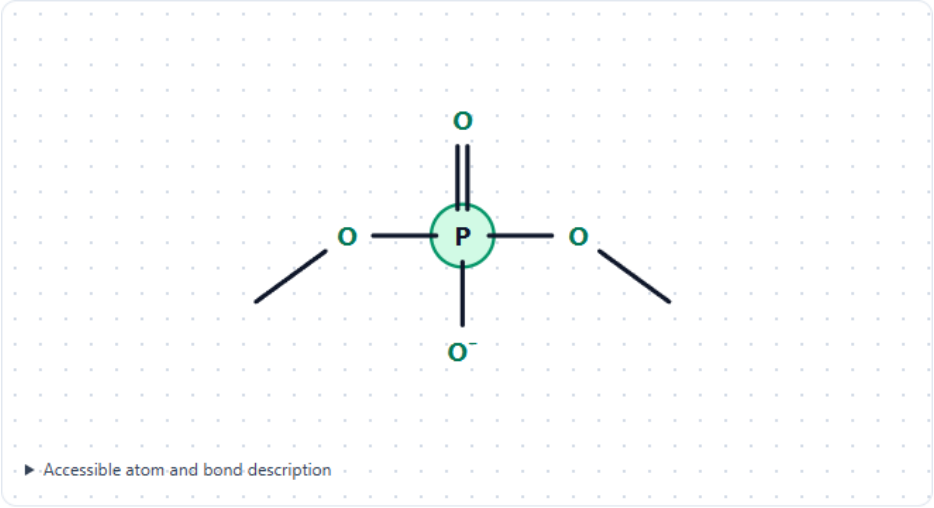
Answer type: choice

Option 0 (A): Phosphodiester

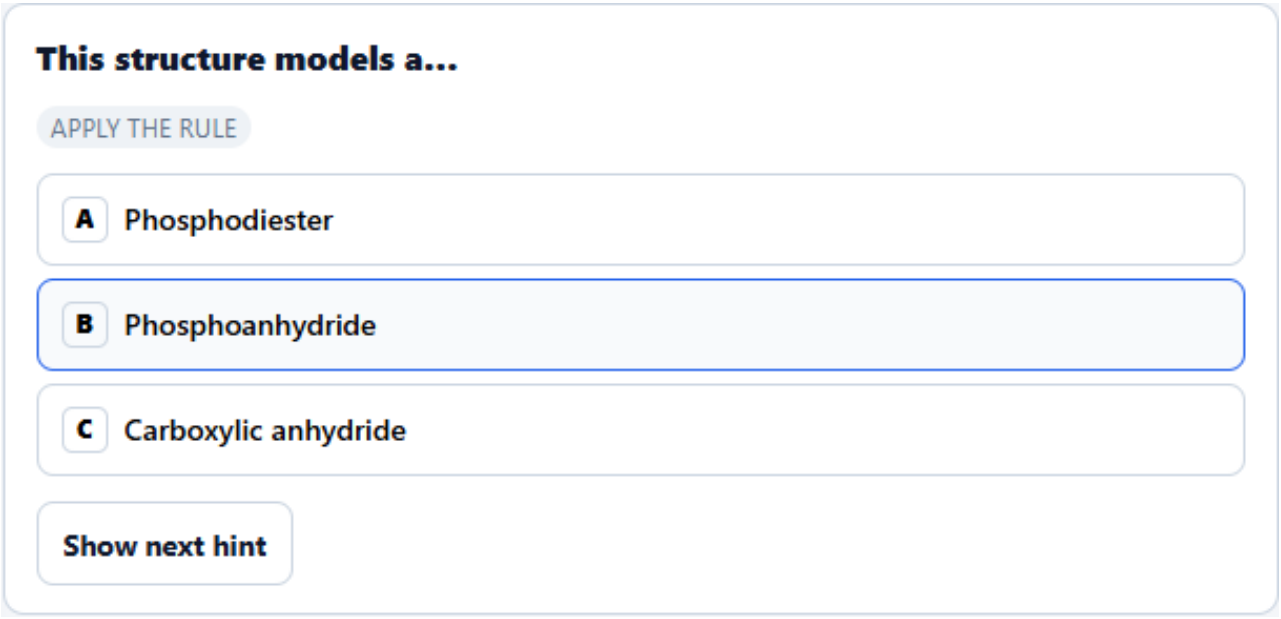
Option 1 (B): Phosphoanhydride

Option 2 (C): Carboxylic anhydride

Screenshot: phosphate-links-stage-2-molecule



Screenshot: phosphate-links-stage-2-question



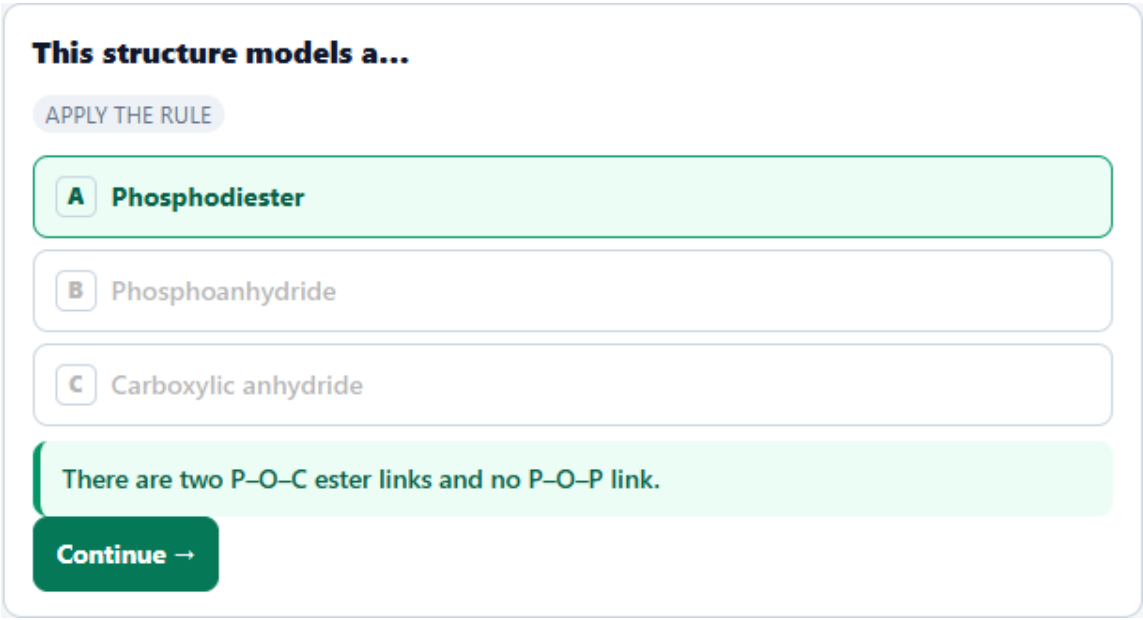

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Count oxygen links to carbon groups.

Hint 2: Only one phosphorus is present.

Explanation: There are two P–O–C ester links and no P–O–P link.

Screenshot: phosphate-links-stage-2-feedback




## phosphate-links:stage-3 | Why can ATP hydrolysis be favorable overall?

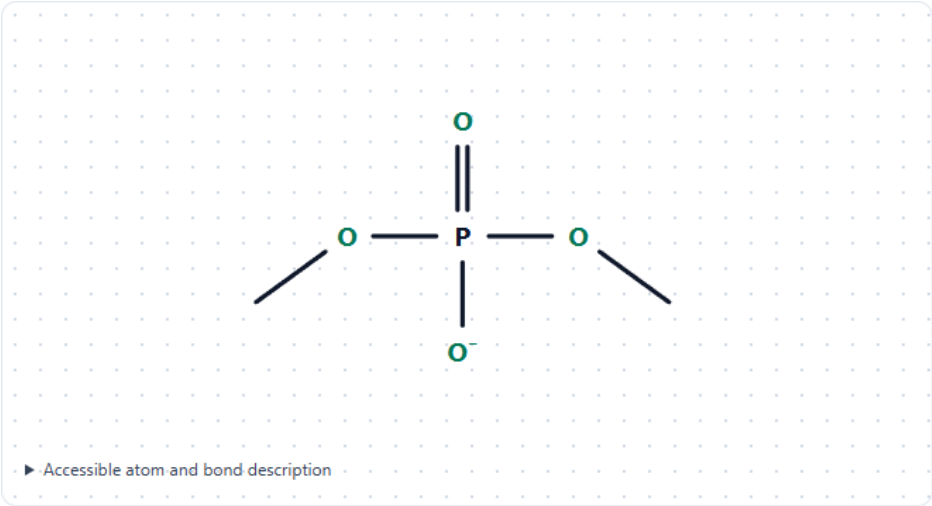
Answer type: choice

Option 0 (A): Products have lower free energy under the stated conditions

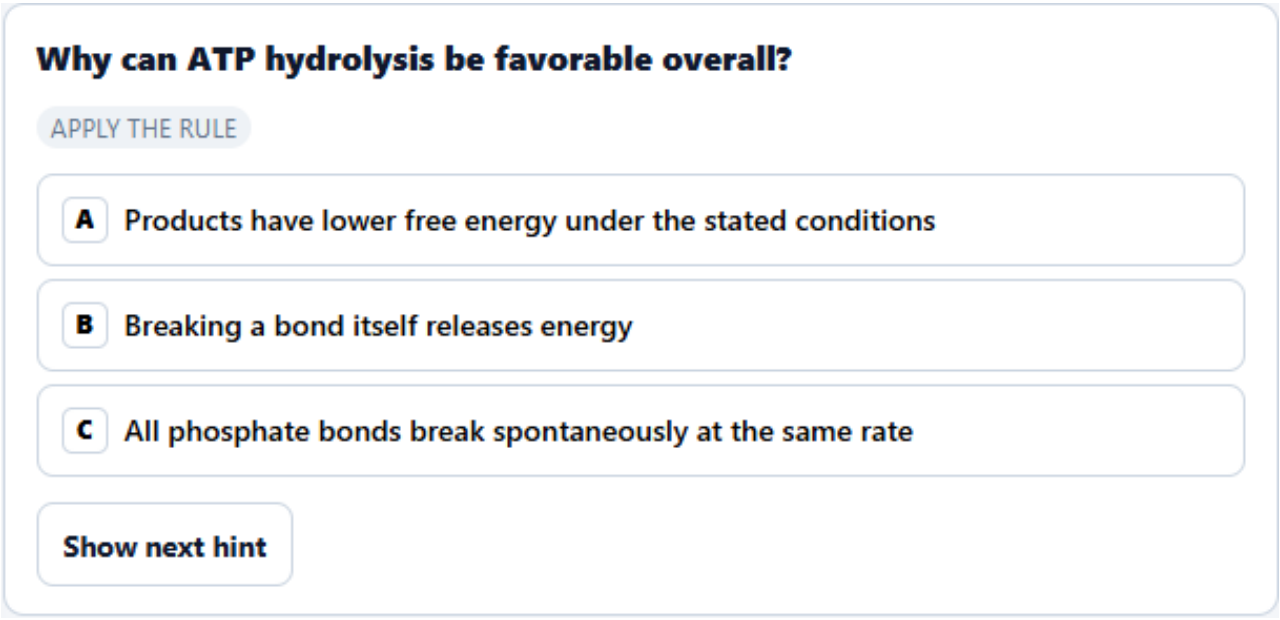
Option 1 (B): Breaking a bond itself releases energy

Option 2 (C): All phosphate bonds break spontaneously at the same rate

Screenshot: phosphate-links-stage-3-molecule




Screenshot: phosphate-links-stage-3-question



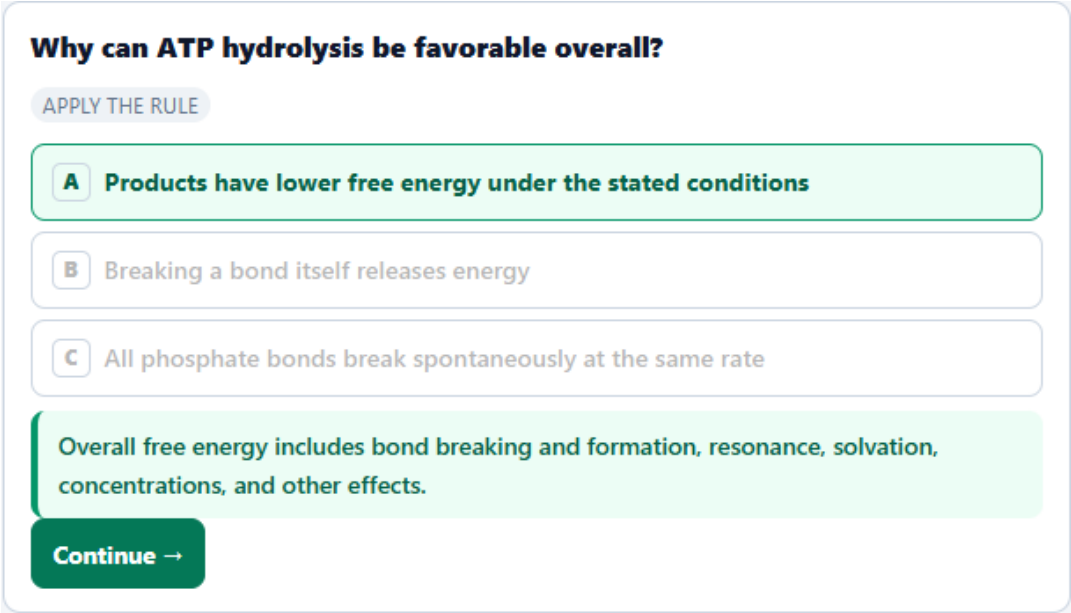

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish a single bond-breaking step from the full reaction.

Hint 2: Thermodynamics does not determine the rate by itself.

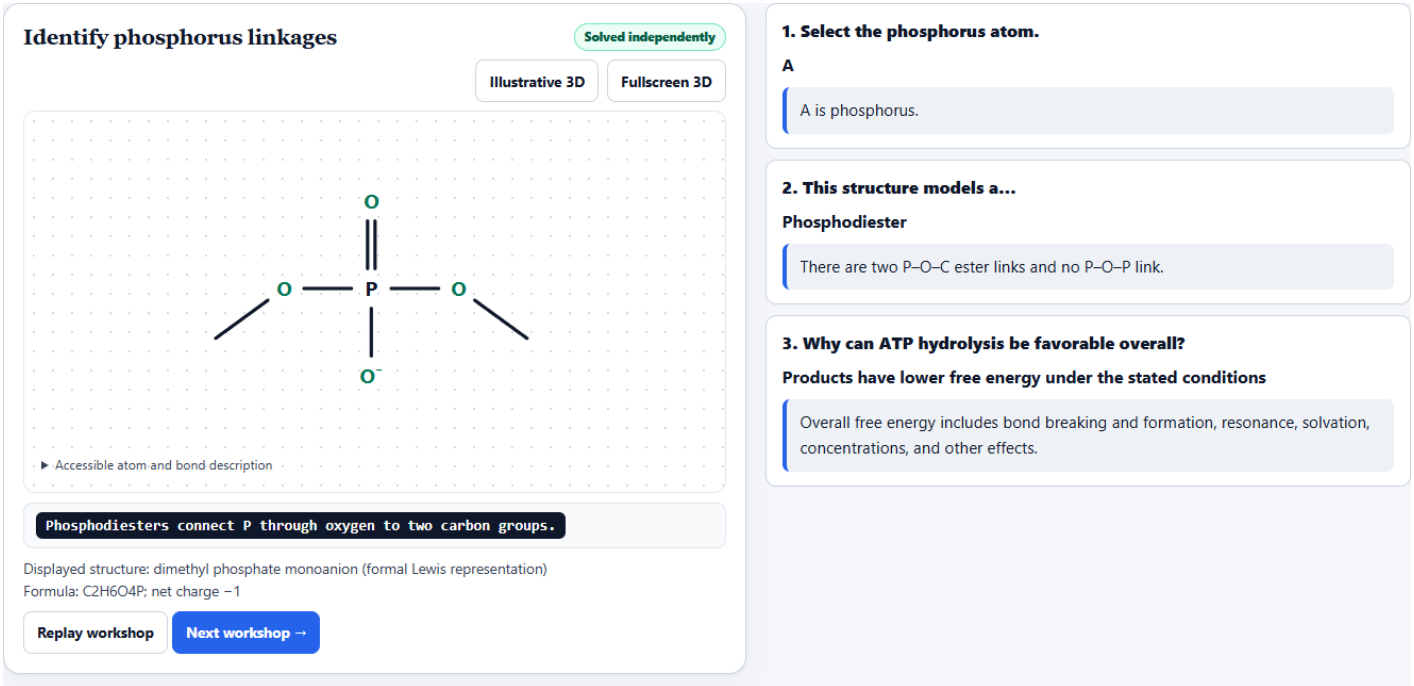
Explanation: Overall free energy includes bond breaking and formation, resonance, solvation, concentrations, and other effects.

Screenshot: phosphate-links-stage-3-feedback




Completion summary: Phosphodiesters connect P through oxygen to two carbon groups.

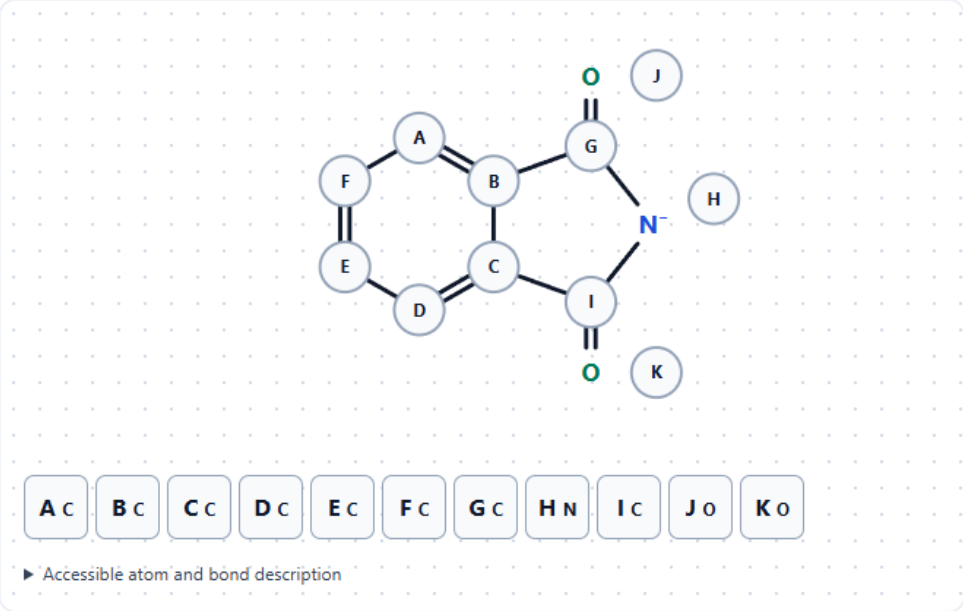
Screenshot: phosphate-links-complete



# gabriel-route | Track protected nitrogen and side-chain installation

The canvas shows the protected nitrogen reagent, with its potassium counterion omitted; later questions refer to the route described in Learn & Explore.

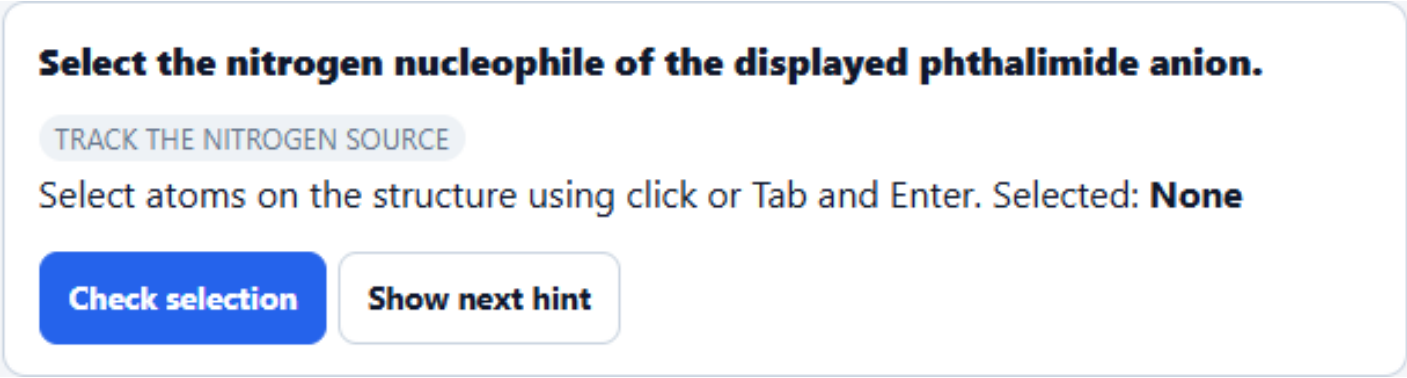
Screenshot: gabriel-route-stage-1-molecule




## gabriel-route:stage-1 | Select the nitrogen nucleophile of the displayed phthalimide anion.

Answer type: atoms

Screenshot: gabriel-route-stage-1-question



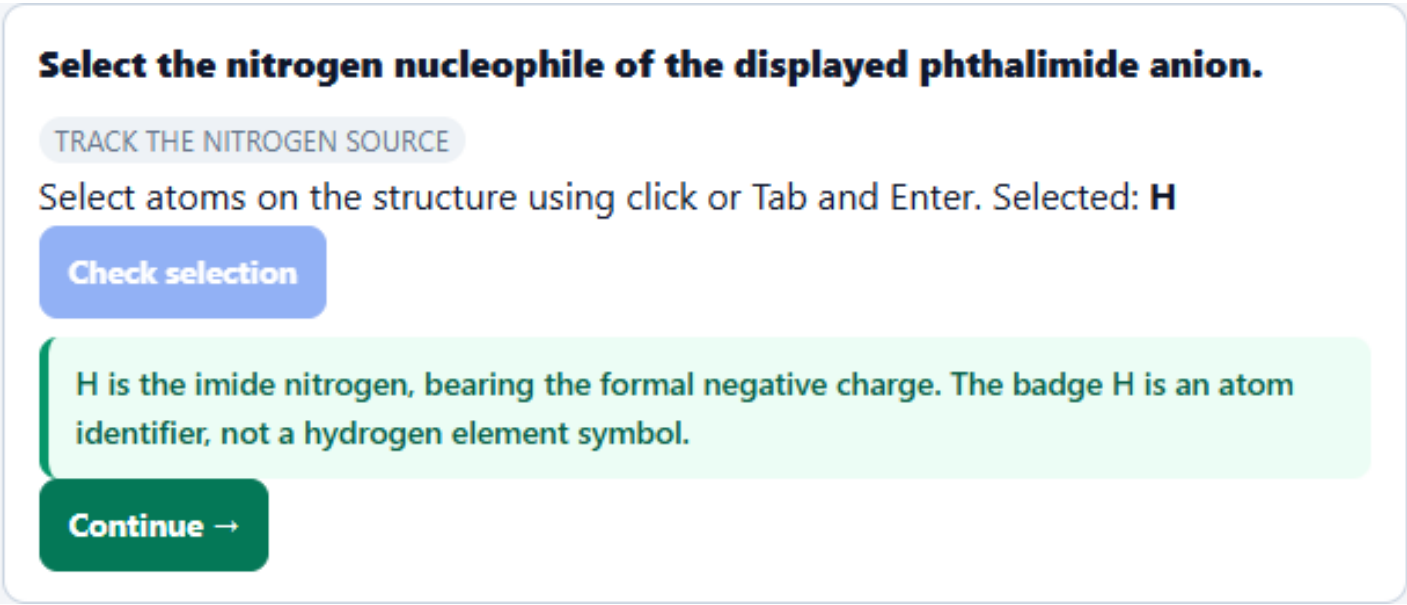

{"authorAnswer": [7], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [7]}}

Hint 1: Locate N in the five-membered imide ring.

Hint 2: The nucleophilic atom bears the formal negative charge.

Explanation: H is the imide nitrogen, bearing the formal negative charge. The badge H is an atom identifier, not a hydrogen element symbol.

Screenshot: gabriel-route-stage-1-feedback




## gabriel-route:stage-2 | After protected aminomalonate formation and deprotonation, a suitable primary alkyl halide installs the side chain mainly by...

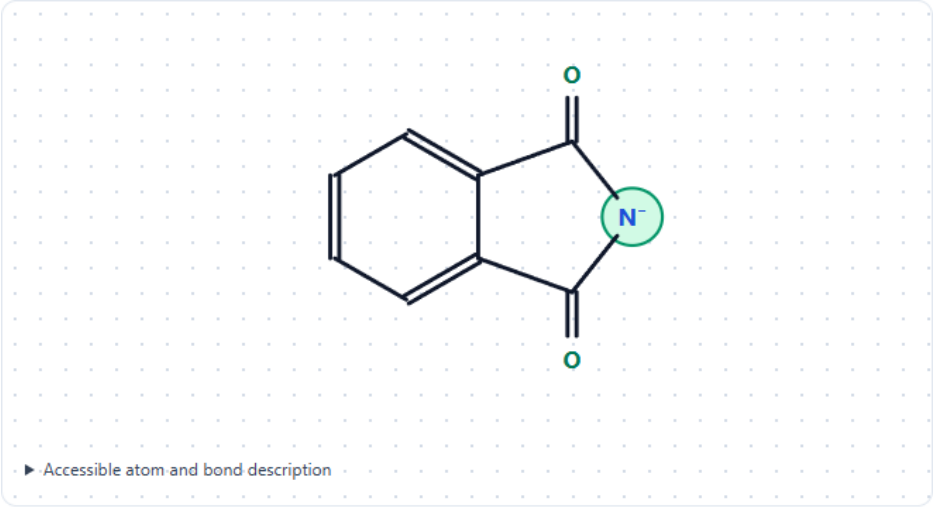
Answer type: choice

Option 0 (A): SN2 carbon alkylation

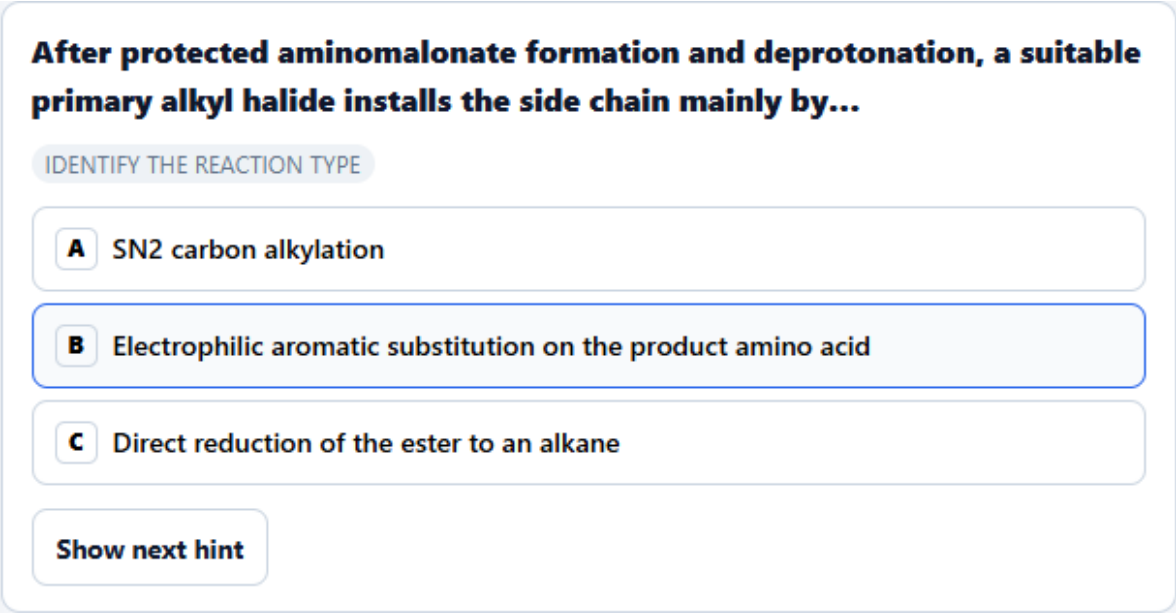
Option 1 (B): Electrophilic aromatic substitution on the product amino acid

Option 2 (C): Direct reduction of the ester to an alkane

Screenshot: gabriel-route-stage-2-molecule



Screenshot: gabriel-route-stage-2-question



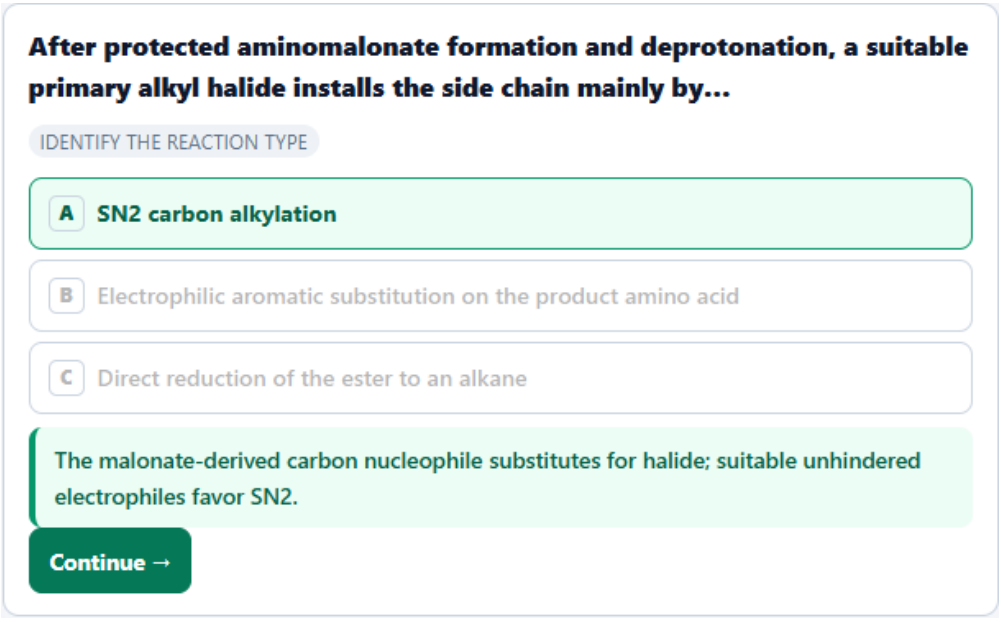

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Identify the nucleophile and the leaving group.

Hint 2: An alkyl halide is attacked at its saturated carbon.

Explanation: The malonate-derived carbon nucleophile substitutes for halide; suitable unhindered electrophiles favor SN2.

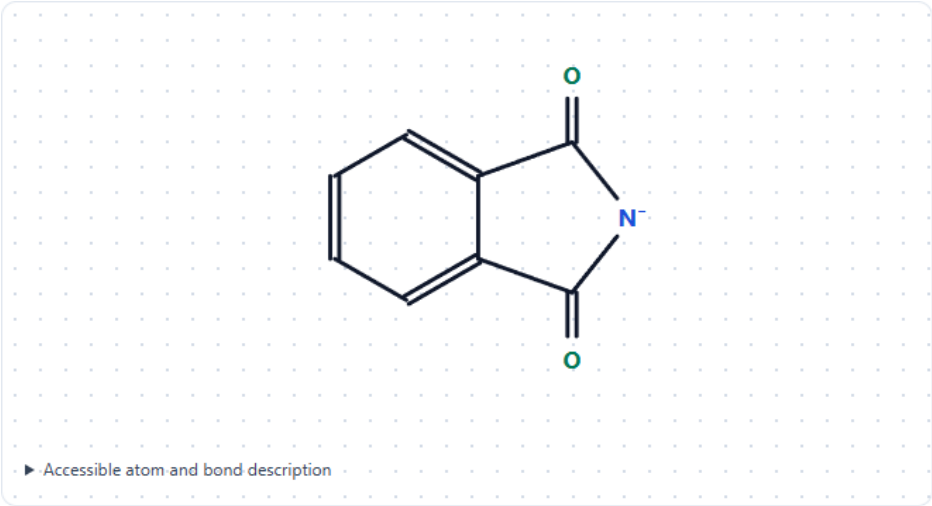
Screenshot: gabriel-route-stage-2-feedback



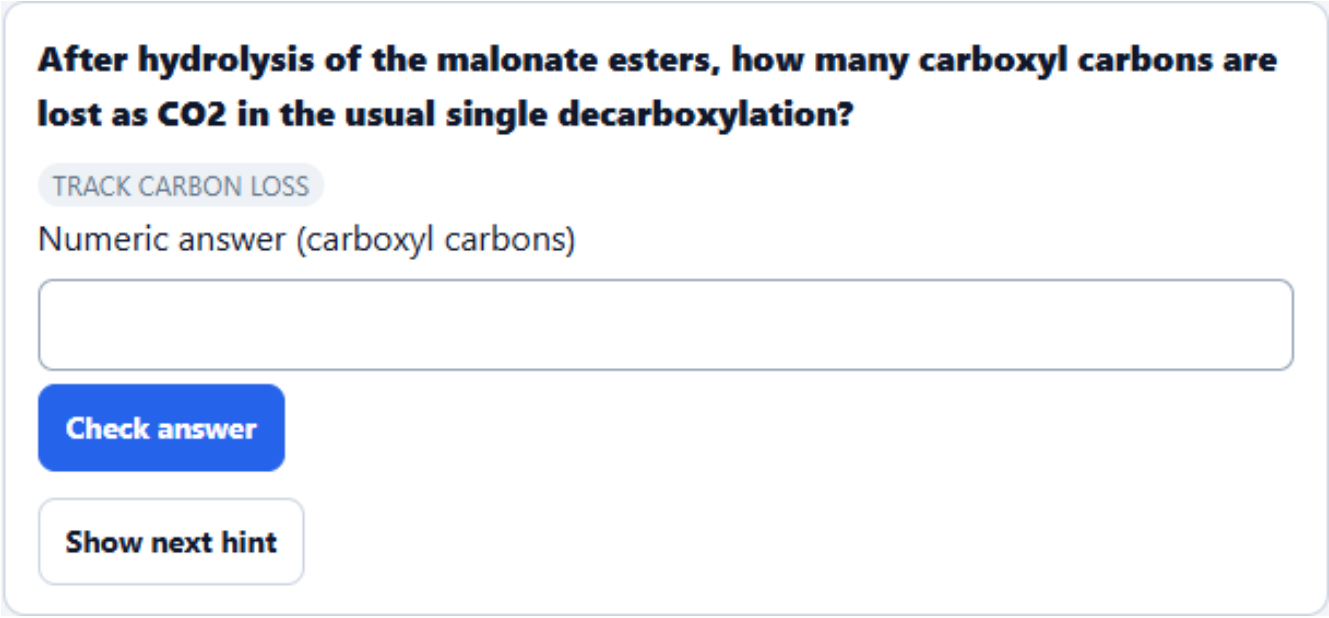

## gabriel-route:stage-3 | After hydrolysis of the malonate esters, how many carboxyl carbons are lost as CO2 in the usual single decarboxylation?

Answer type: number | unit: carboxyl carbons

Screenshot: gabriel-route-stage-3-molecule




Screenshot: gabriel-route-stage-3-question



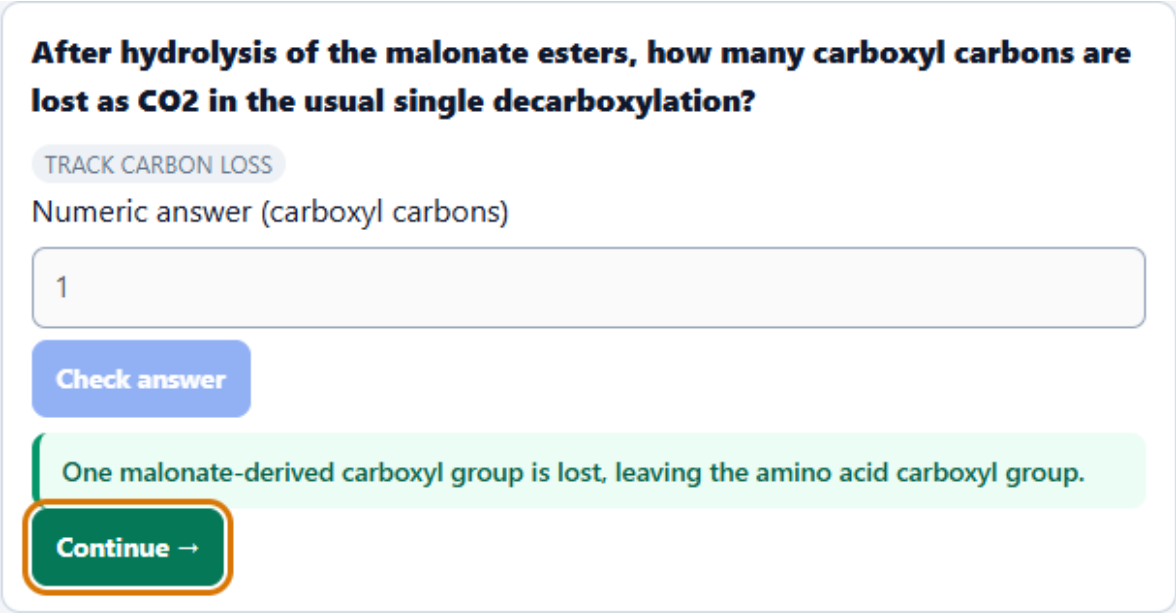

{"authorAnswer": 1, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish deprotection from the decarboxylation step.

Hint 2: The aminomalonic acid framework starts with two carboxyl groups and retains one.

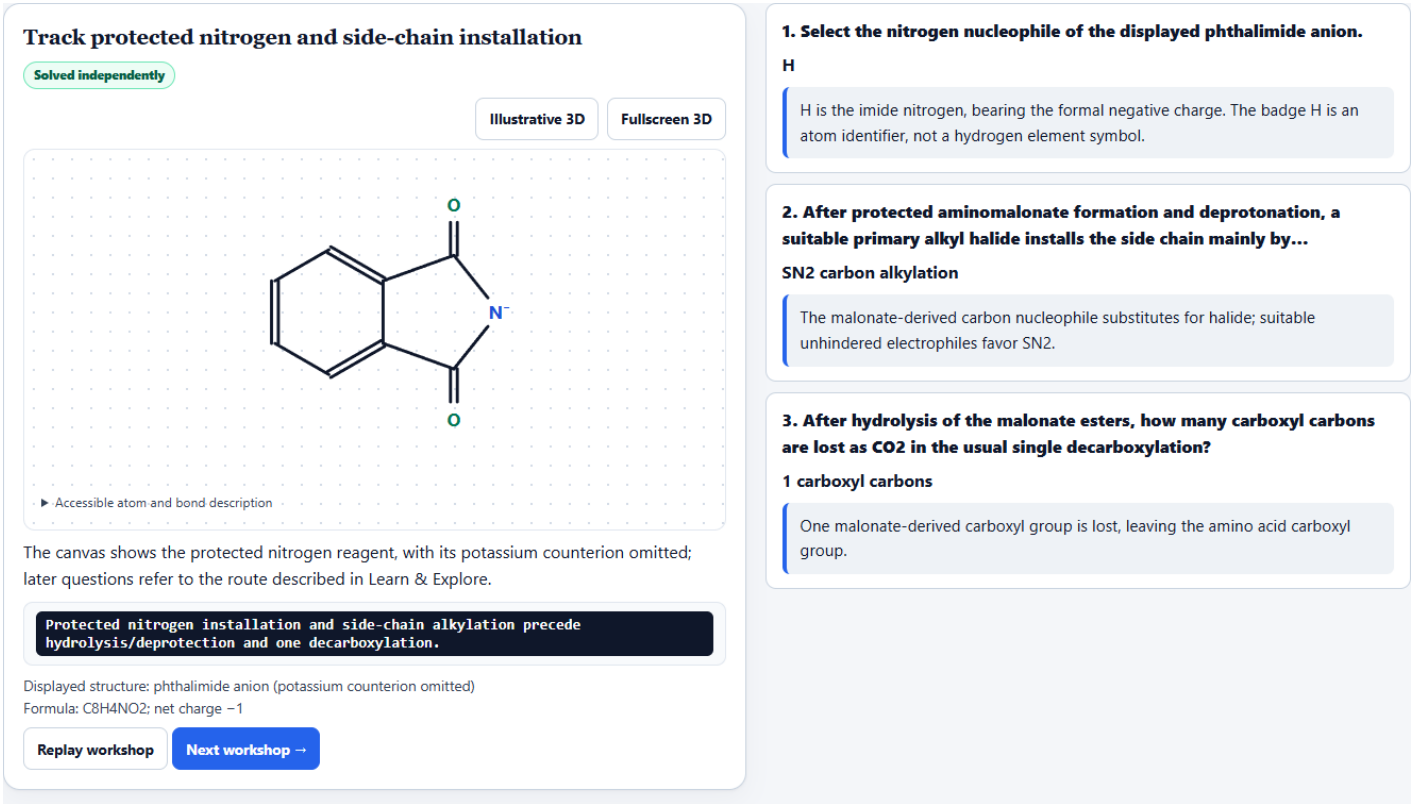
Explanation: One malonate-derived carboxyl group is lost, leaving the amino acid carboxyl group.

Screenshot: gabriel-route-stage-3-feedback




Completion summary: Protected nitrogen installation and side-chain alkylation precede hydrolysis/deprotection and one decarboxylation.

Screenshot: gabriel-route-complete



# Molecular repository: author claims to verify

## alanine

Author claims: alanine zwitterion (stereochemistry unspecified) | formula C3H7NO2 | canonical composition C3H7NO2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=45.00, y=115.00; charge=1; explicit H override=none

1 (B): C; x=105.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=165.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=165.00, y=175.00; charge=0; explicit H override=none

4 (E): O; x=225.00, y=85.00; charge=-1; explicit H override=none

5 (F): C; x=105.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [1, 5, 1]]

## glycine

Author claims: glycine zwitterion | formula C2H5NO2 | canonical composition C2H5NO2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=45.00, y=115.00; charge=1; explicit H override=none

1 (B): C; x=105.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=165.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=165.00, y=175.00; charge=0; explicit H override=none

4 (E): O; x=225.00, y=85.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1]]

## glycylglycine

Author claims: glycylglycine zwitterion | formula C4H8N2O3 | canonical composition C4H8N2O3 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=35.00, y=110.00; charge=1; explicit H override=none

1 (B): C; x=90.00, y=80.00; charge=0; explicit H override=none

2 (C): C; x=145.00, y=110.00; charge=0; explicit H override=none

3 (D): O; x=145.00, y=170.00; charge=0; explicit H override=none

4 (E): N; x=200.00, y=80.00; charge=0; explicit H override=none

5 (F): C; x=255.00, y=110.00; charge=0; explicit H override=none

6 (G): C; x=310.00, y=80.00; charge=0; explicit H override=none

7 (H): O; x=310.00, y=20.00; charge=0; explicit H override=none

8 (I): O; x=365.00, y=110.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [5, 6, 1], [6, 7, 2], [6, 8, 1]]

## aminonitrile

Author claims: 2-aminopropanenitrile (stereochemistry unspecified) | formula C3H6N2 | canonical composition C3H6N2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=110.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=170.00, y=120.00; charge=0; explicit H override=none

3 (D): N; x=230.00, y=155.00; charge=0; explicit H override=none

4 (E): N; x=110.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 3], [1, 4, 1]]

## phthalimide-anion

Author claims: phthalimide anion (potassium counterion omitted) | formula C8H4NO2 | canonical composition C8H4NO2 | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=130.00, y=45.00; charge=0; explicit H override=none

1 (B): C; x=177.63, y=72.50; charge=0; explicit H override=none

2 (C): C; x=177.63, y=127.50; charge=0; explicit H override=none

3 (D): C; x=130.00, y=155.00; charge=0; explicit H override=none

4 (E): C; x=82.37, y=127.50; charge=0; explicit H override=none

5 (F): C; x=82.37, y=72.50; charge=0; explicit H override=none

6 (G): C; x=240.00, y=50.00; charge=0; explicit H override=none

7 (H): N; x=280.00, y=100.00; charge=-1; explicit H override=0

8 (I): C; x=240.00, y=150.00; charge=0; explicit H override=none

9 (J): O; x=240.00, y=5.00; charge=0; explicit H override=none

10 (K): O; x=240.00, y=195.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2], [3, 4, 1], [4, 5, 2], [5, 0, 1], [1, 6, 1], [6, 7, 1], [7, 8, 1], [8, 2, 1], [6, 9, 2], [8, 10, 2]]

## dimethylphosphate

Author claims: dimethyl phosphate monoanion (formal Lewis representation) | formula C2H6O4P | canonical composition C2H6O4P | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): P; x=160.00, y=100.00; charge=0; explicit H override=none

1 (B): O; x=160.00, y=30.00; charge=0; explicit H override=none

2 (C): O; x=90.00, y=100.00; charge=0; explicit H override=none

3 (D): O; x=230.00, y=100.00; charge=0; explicit H override=none

4 (E): O; x=160.00, y=170.00; charge=-1; explicit H override=none

5 (F): C; x=35.00, y=140.00; charge=0; explicit H override=none

6 (G): C; x=285.00, y=140.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [0, 2, 1], [0, 3, 1], [0, 4, 1], [2, 5, 1], [3, 6, 1]]

## phosphoricacid

Author claims: phosphoric acid (formal Lewis representation) | formula H3PO4 | canonical composition H3O4P | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): P; x=150.00, y=100.00; charge=0; explicit H override=none

1 (B): O; x=150.00, y=30.00; charge=0; explicit H override=none

2 (C): O; x=80.00, y=100.00; charge=0; explicit H override=none

3 (D): O; x=220.00, y=100.00; charge=0; explicit H override=none

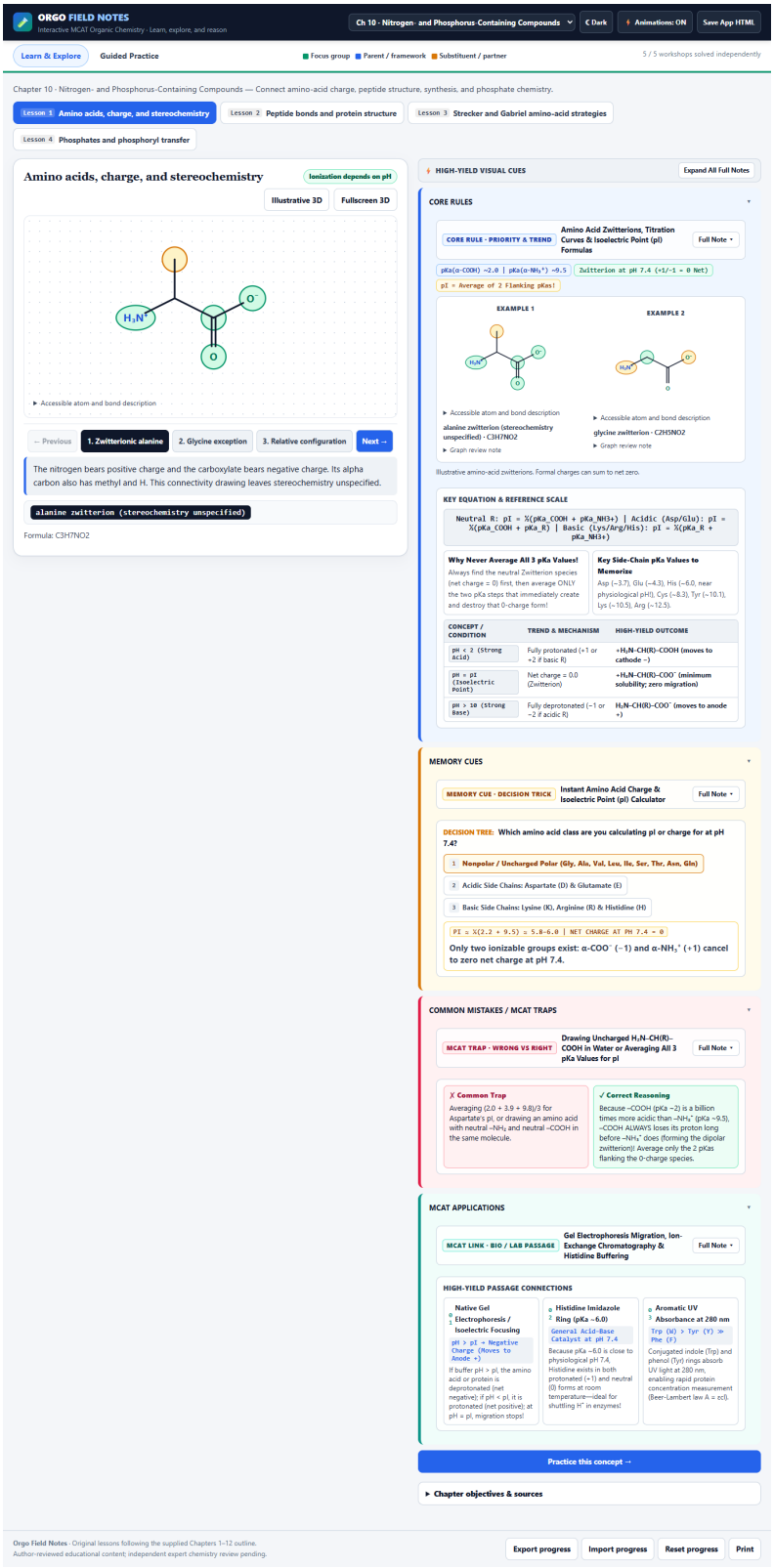
4 (E): O; x=150.00, y=170.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [0, 2, 1], [0, 3, 1], [0, 4, 1]]

# Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

Screenshot: layout-desktop-light



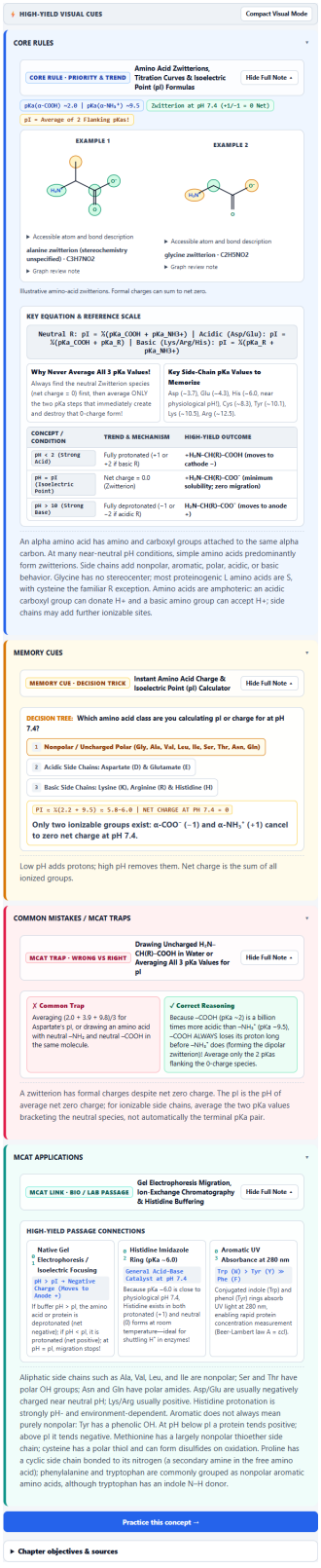
Screenshot: layout-mobile-light



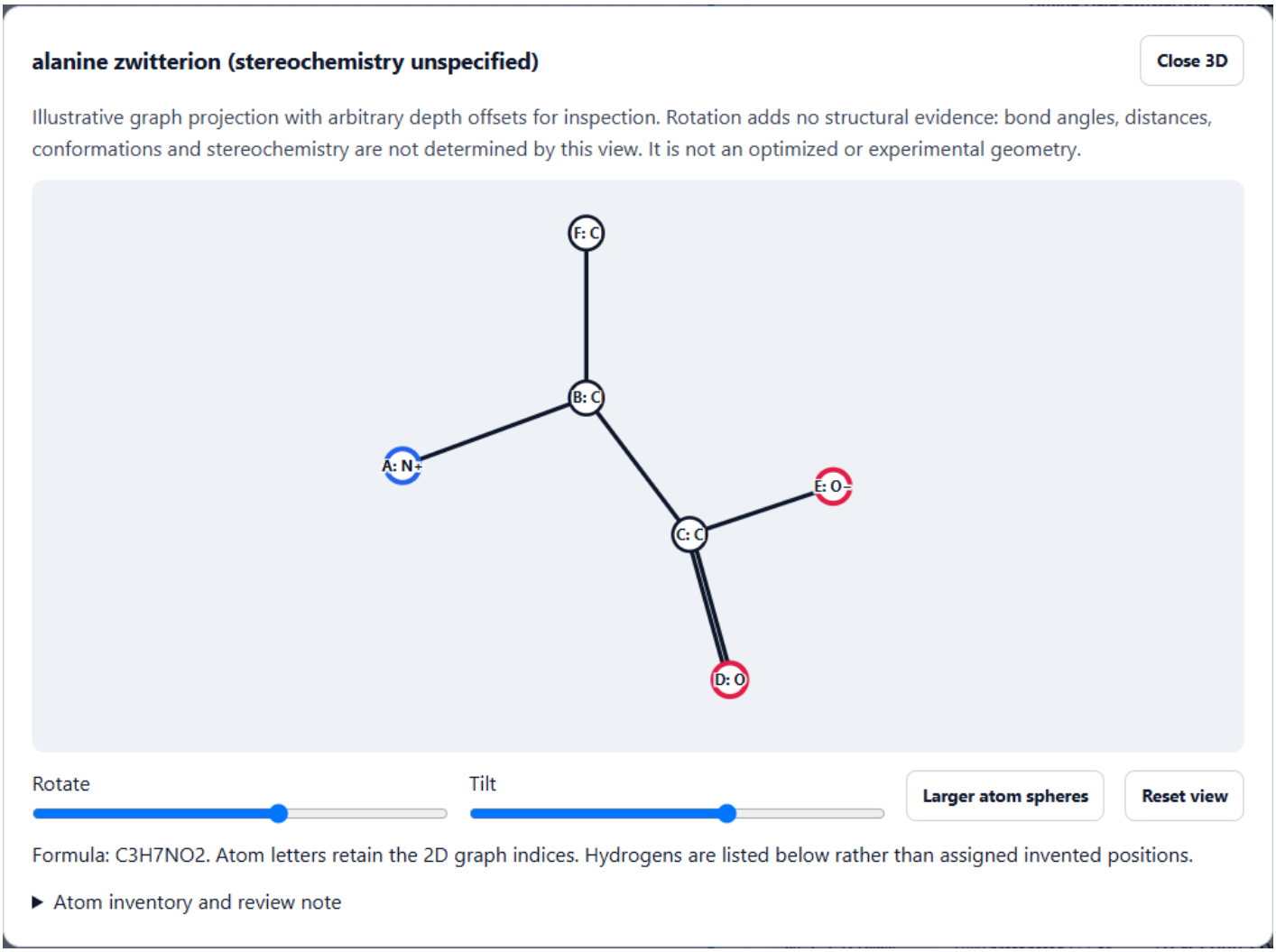
Screenshot: layout-mobile-dark



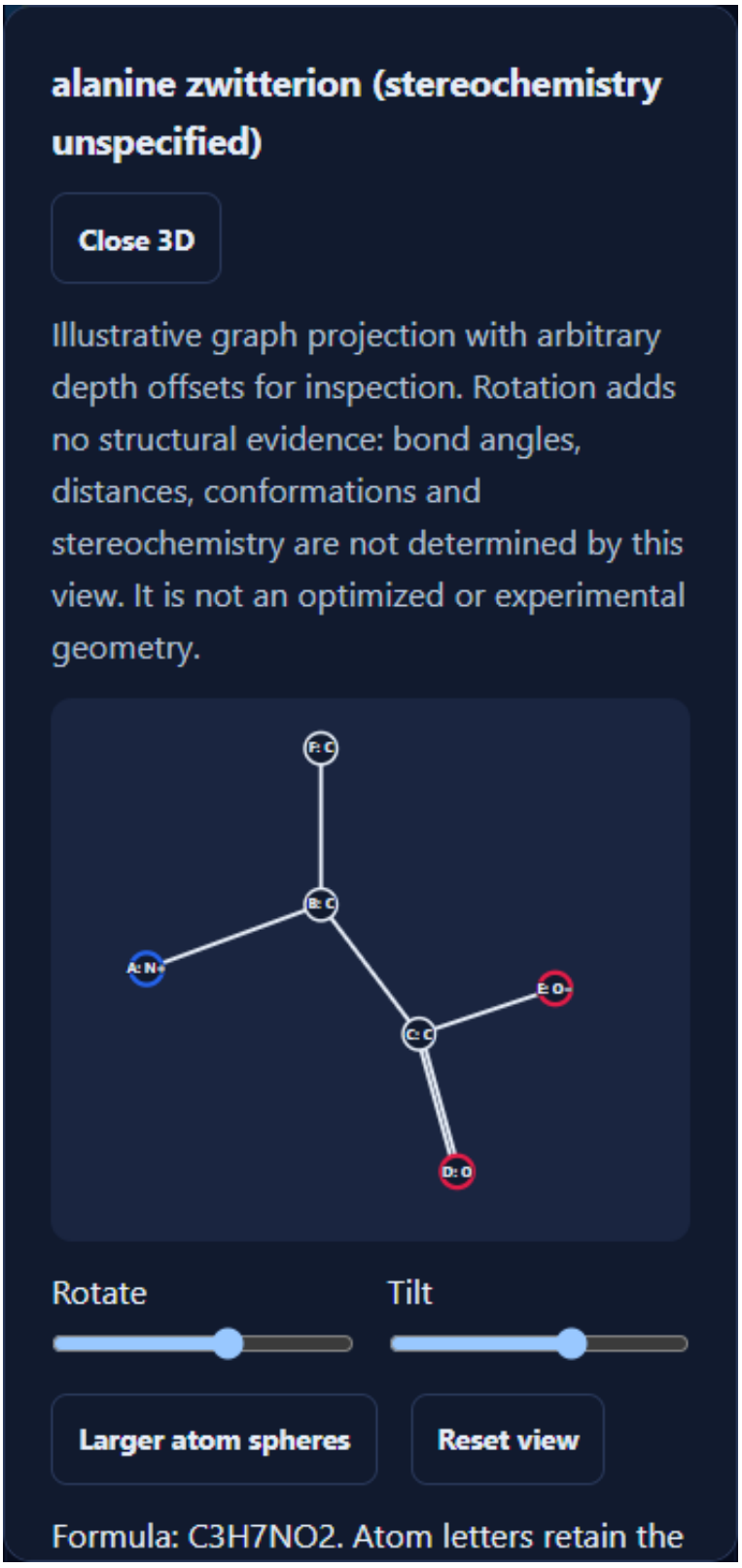
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark




# Execution evidence

72 screenshots and 16 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.